RAG EVAL 2.0

Multimodal Document Ingestion and Retrieval Testing Report

ICS 499 Capstone | October 20, 2026

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Repository: https://github.com/sjasthi/llm-rag-evals

Executive summary

This report documents ten controlled input test cases used to validate the RAG EVAL 2.0 multimodal ingestion, retrieval, and grounded-answer workflow. The tests cover images, PDFs, Markdown, PowerPoint, and Word documents. Evidence is drawn from browser ingestion and chat screenshots, MySQL metadata checks, and the expected responses recorded during testing.

Test Case 6 prompted two retrieval enhancements: natural-language enrichment of table chunks and a content-aware reranker. Final browser validation confirmed accurate table and diagram answers using Hybrid RRF with only three retrieved sources. Test Case 8 also documents earlier Mermaid-specific retrieval improvements using hybrid reciprocal rank fusion.

System and validation method

Ingestion: Files are uploaded through the browser and processed into typed document elements. Native text and tables are extracted directly where available; visual content is described by a vision-language model. Content and provenance are saved in MySQL and indexed in ChromaDB.

Retrieval and answering: Questions use ChromaDB vector search, MySQL keyword search, or Hybrid Reciprocal Rank Fusion (RRF). The updated pipeline can rerank candidates based on question intent and evidence overlap. Retrieved chunks are supplied to the configured Gemini answer model to produce grounded responses.

Validation criteria: (1) successful upload and ingestion, (2) appropriate stored content and provenance, (3) relevant retrieved evidence, and (4) an answer matching the controlled expected result. Screenshots are retained as the primary test evidence. The report distinguishes confirmed passes from observations requiring further investigation.

Test fixture index

TC01 — Standalone image (PNG/JPEG): View fixture on GitHub

TC02 — PDF containing an embedded image: View fixture on GitHub

TC03 — PDF containing a native table: View fixture on GitHub

TC04 — Text-only PDF: View fixture on GitHub

TC05 — PDF with text, table, and image: View fixture on GitHub

TC06 — PDF with two text sections, two tables, and two images: View fixture on GitHub

TC07 — Formatted Markdown: View fixture on GitHub

TC08 — Markdown with Mermaid diagram: View fixture on GitHub

TC09 — PowerPoint with text, tables, and images: View fixture on GitHub

TC10 — Word document with text, tables, and images: View fixture on GitHub

Results overview

Case	Input	Documented outcome
TC01	Standalone image (PNG/JPEG)	Pass — screenshots supplied
TC02	PDF containing an embedded image	Pass — screenshots supplied
TC03	PDF containing a native table	Pass — screenshots supplied
TC04	Text-only PDF	Pass — duplicate-text observation noted
TC05	PDF with text, table, and image	Pass — screenshots supplied
TC06	PDF with two text sections, two tables, and two images	Pass — screenshots supplied
TC07	Formatted Markdown	Pass — screenshots supplied
TC08	Markdown with Mermaid diagram	Pass — screenshots supplied
TC09	PowerPoint with text, tables, and images	Pass — screenshots supplied
TC10	Word document with text, tables, and images	Pass — screenshots supplied

Results reflect the screenshots and notes in the supplied evidence document; the report does not claim a separate fresh execution of every test.

Test Case 01 — Standalone image (PNG/JPEG)

Input fixture: TC01_image_only.png | Open / download from GitHub

Original test notes and visual evidence

Ingestion

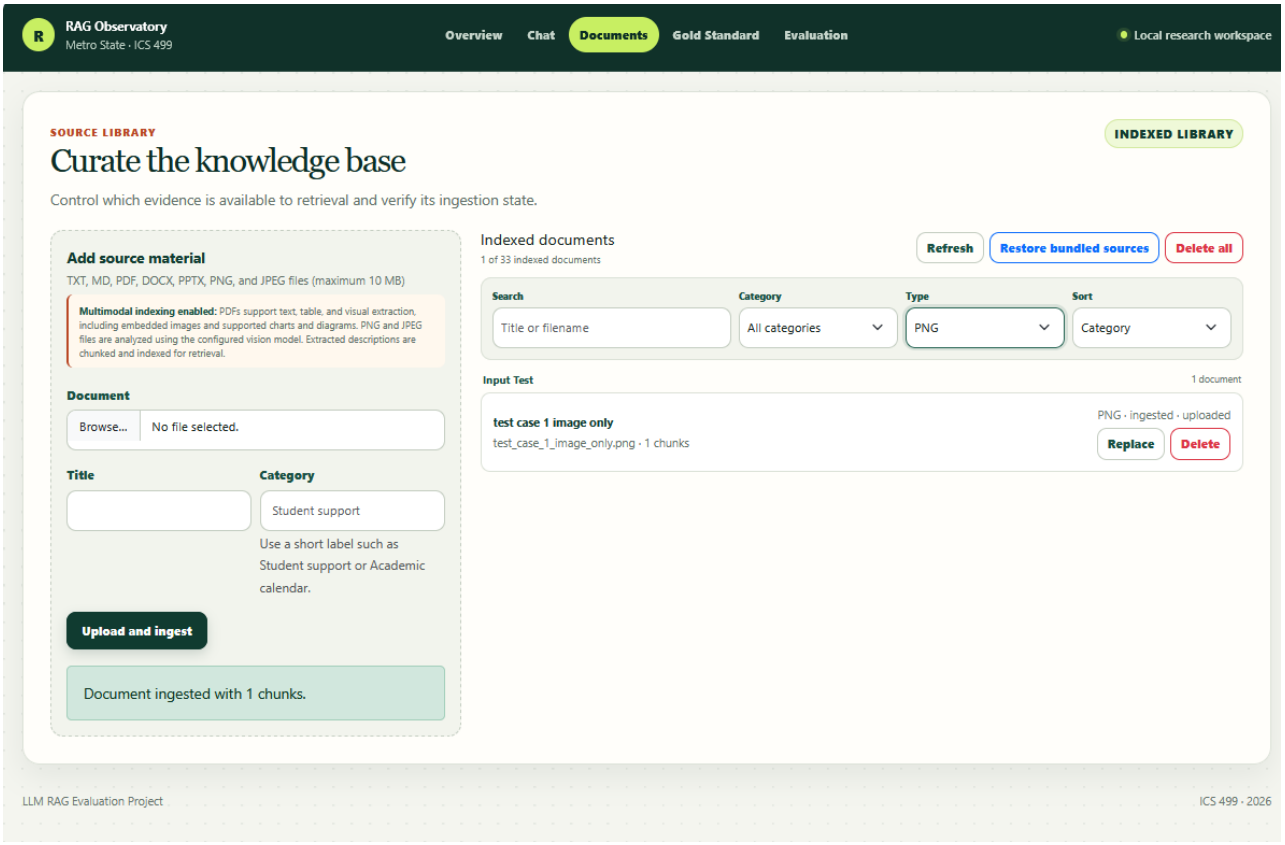


Figure 1.1 — Test Case 1 Document Ingestion

MySQL Metadata Check

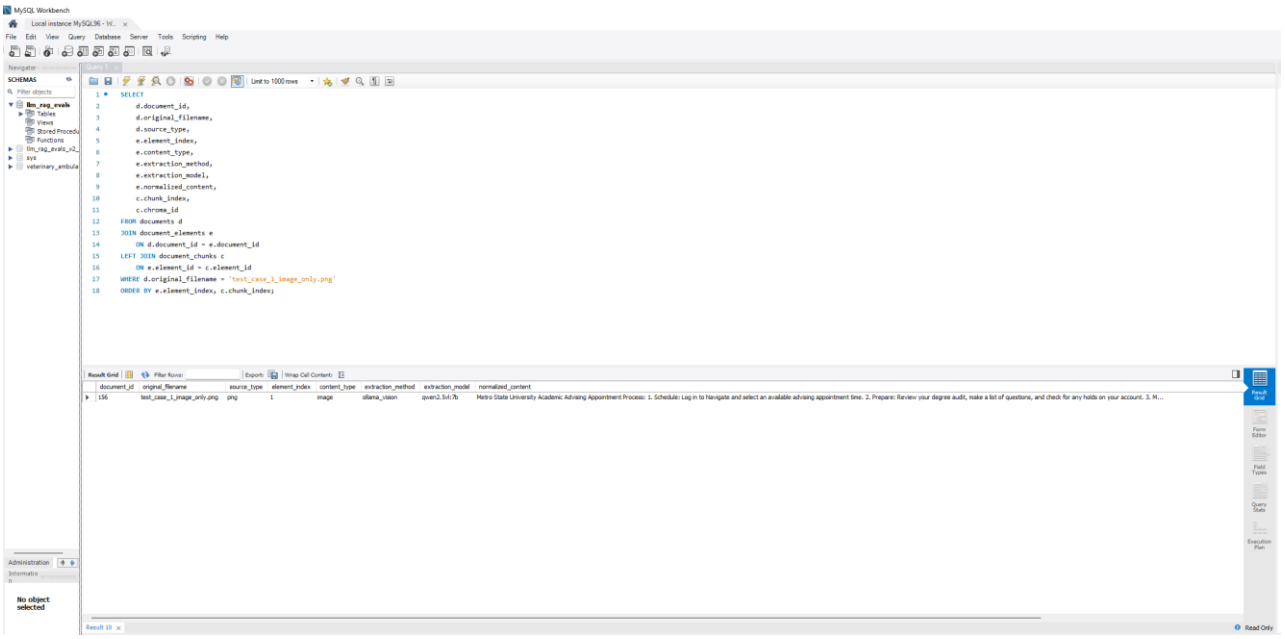


Figure 1.2 — Test Case 1 Metadata Check Screenshot

Normalized_content (description of image)

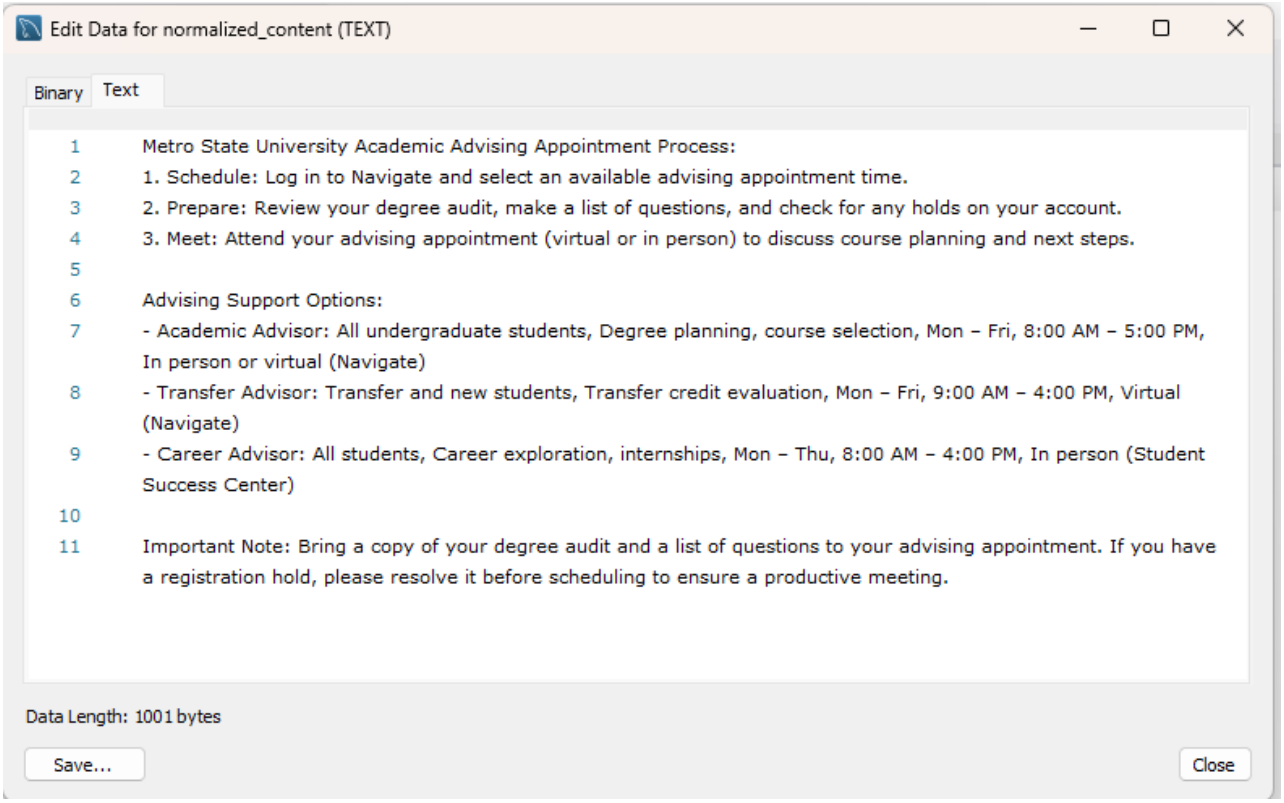


Figure 1.3 — Test Case 1 Image Chunk Description Screenshot

Chat

Process steps: According to the Academic Advising Appointment Process infographic, what are the three steps for scheduling and preparing for an advising appointment?

Expected: Schedule → Prepare → Meet.

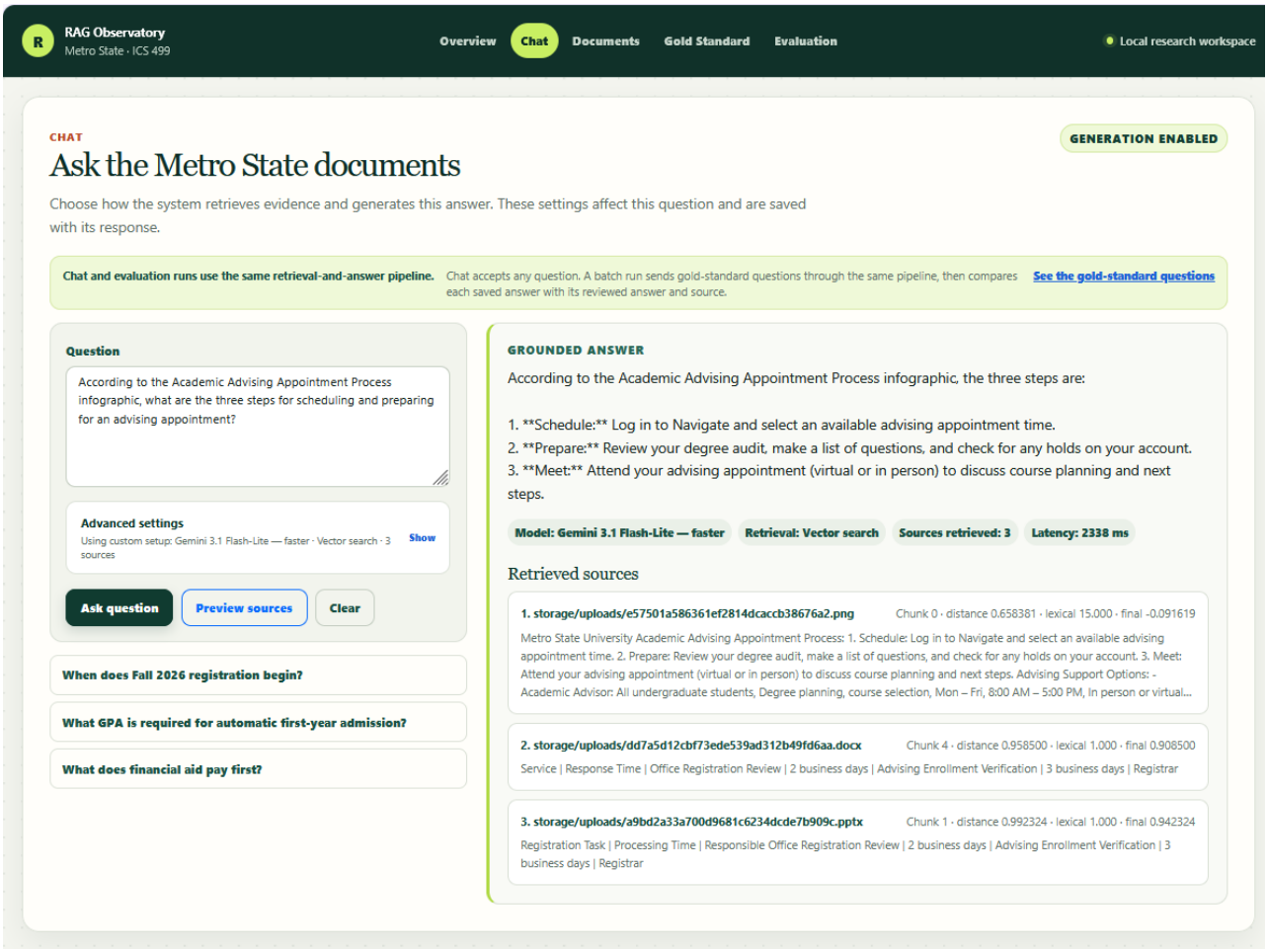


Figure 1.4 — Test Case 1 Question 1 Answer Screenshot

Image table: In the Academic Advising Appointment Process infographic, what are the Career Advisor's availability and appointment location?

Expected: Monday–Thursday, 8:00 AM–4:00 PM; Student Success Center.

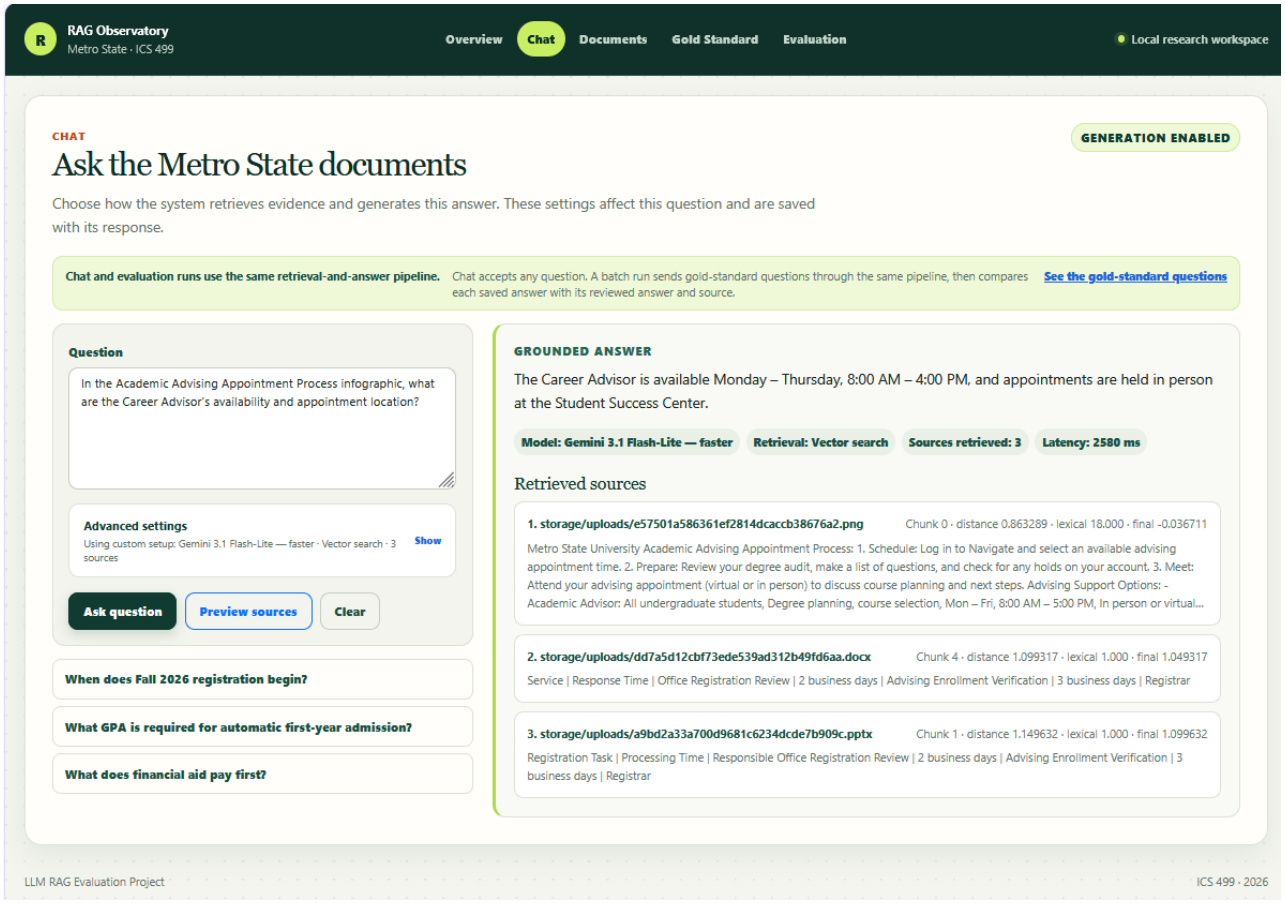


Figure 1.5 — Test Case 1 Question 2 Answer Screenshot

Preparation instructions: According to the Academic Advising Appointment Process infographic, what should a student prepare before meeting with an advisor?

Expected: Review the degree audit, prepare questions, check for account holds, and resolve registration holds before scheduling.

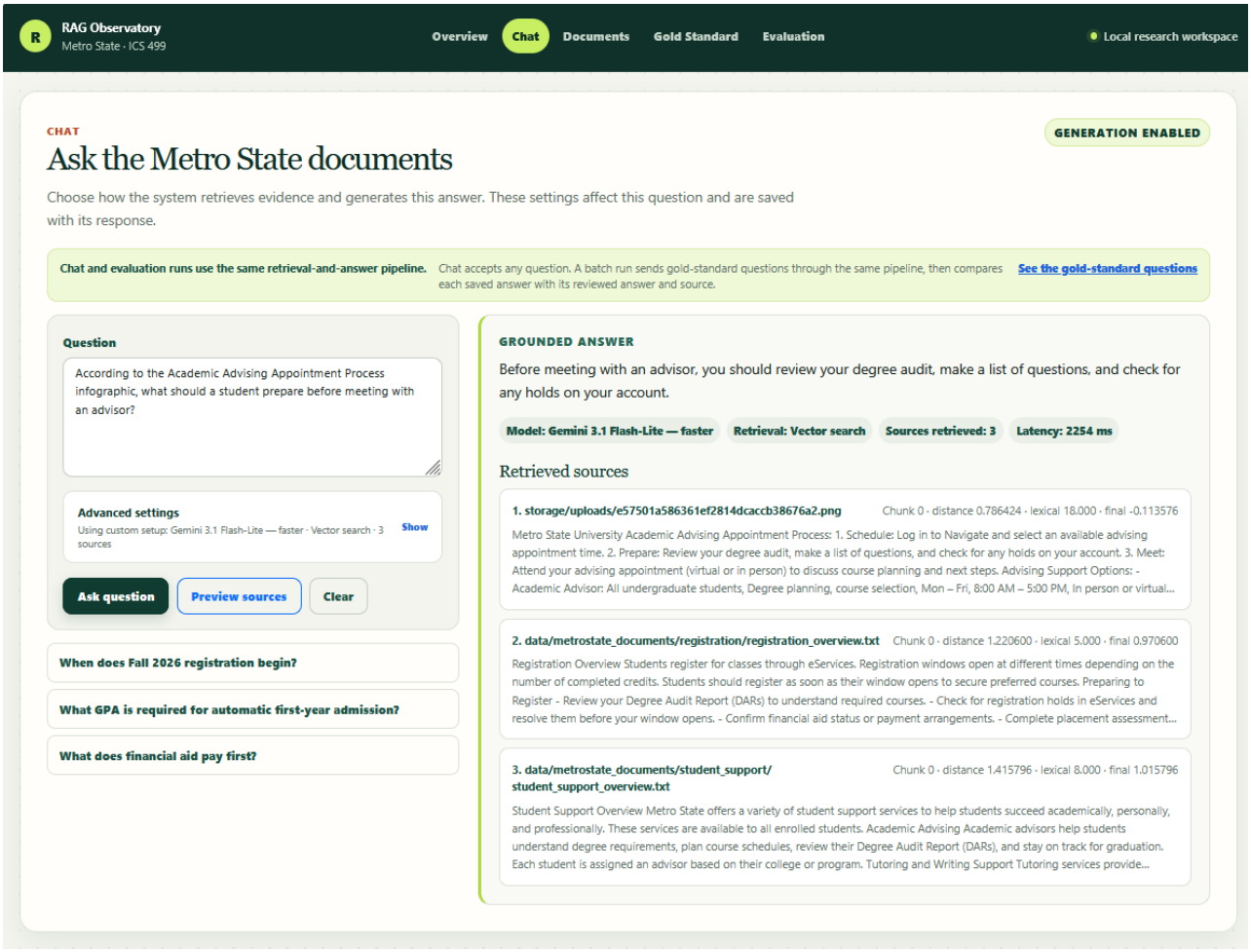


Figure 1.6 — Test Case 1 Question 3 Answer Screenshot

PASS — supplied screenshots document ingestion and grounded answers matching the stated expected results. See individual figures and question/expected-answer notes above.

Test Case 02 — PDF containing an embedded image

Input fixture: TC02_Campus_Water_Savings_Embedded_Image_v3.pdf | Open / download from GitHub

Original test notes and visual evidence

Ingestion

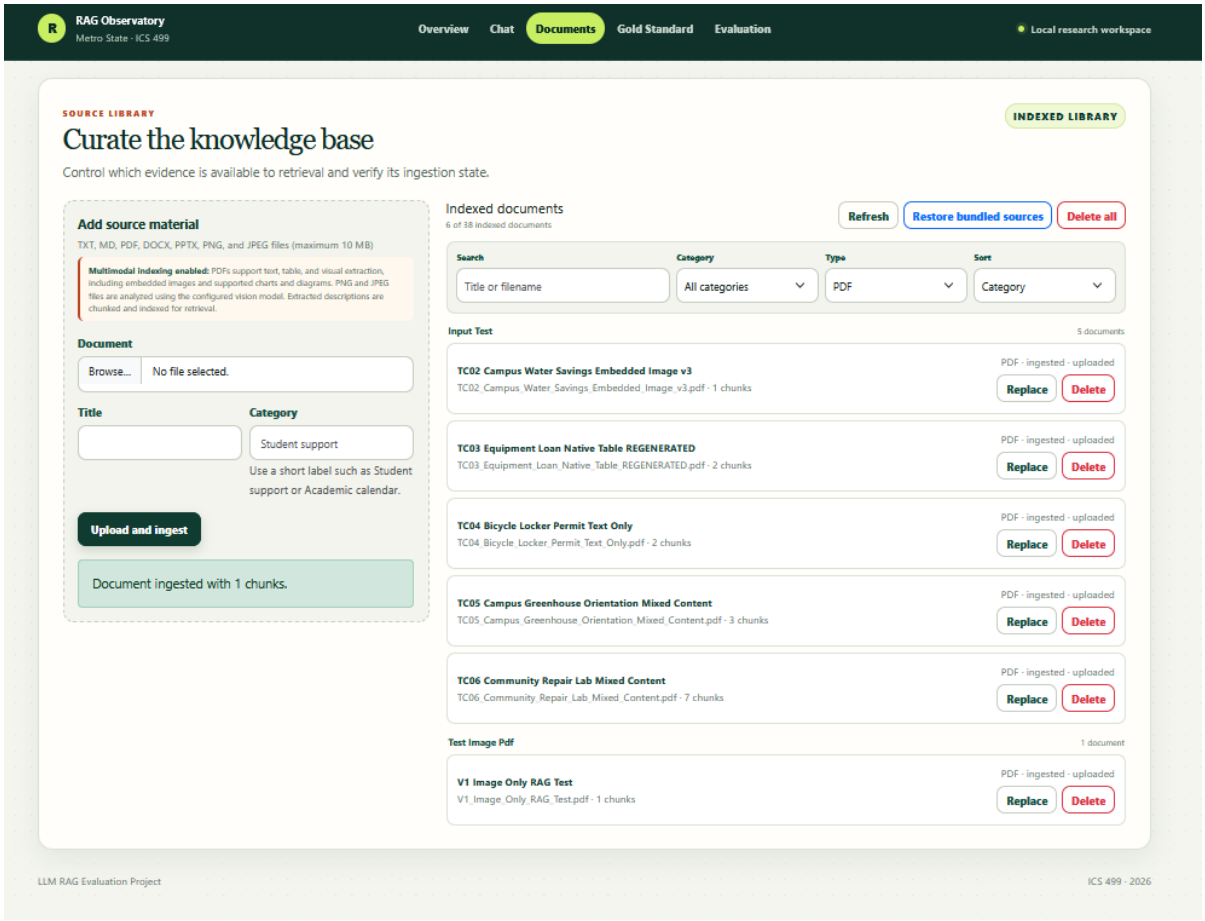


Figure 2.1 — Test Case 2 Document Ingestion Screenshot

MySQL Metadata Check

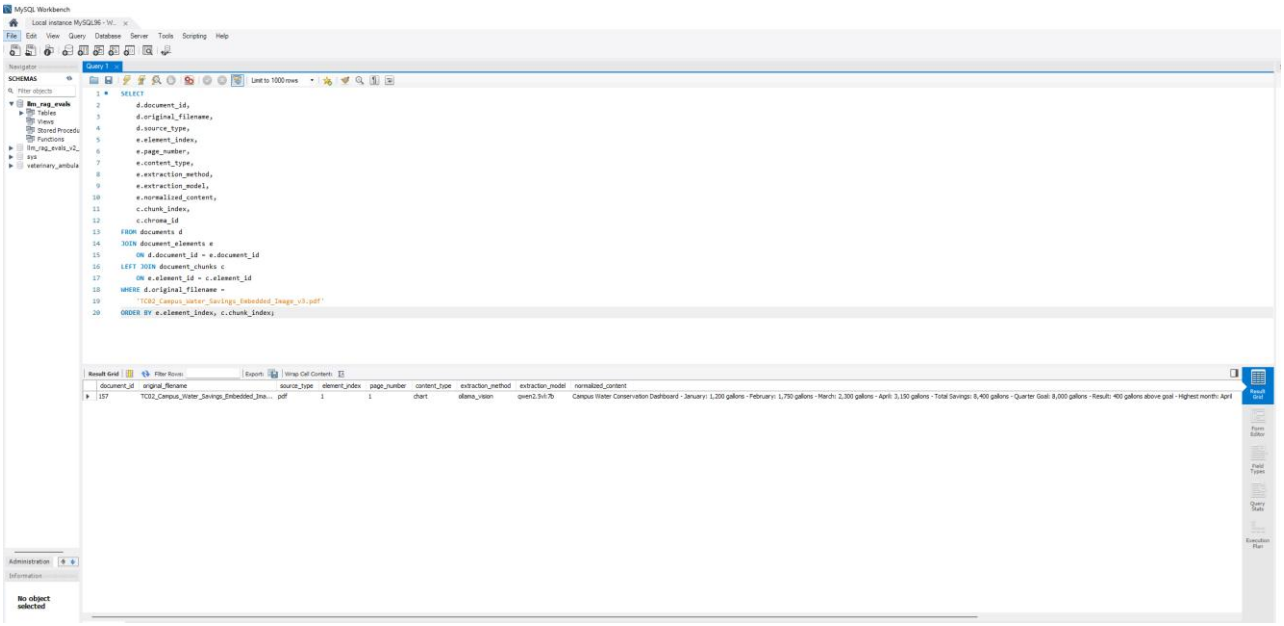


Figure 2.2 — Test Case 2 MySQL Metadata Check Screenshot

Normalized_content (image description)

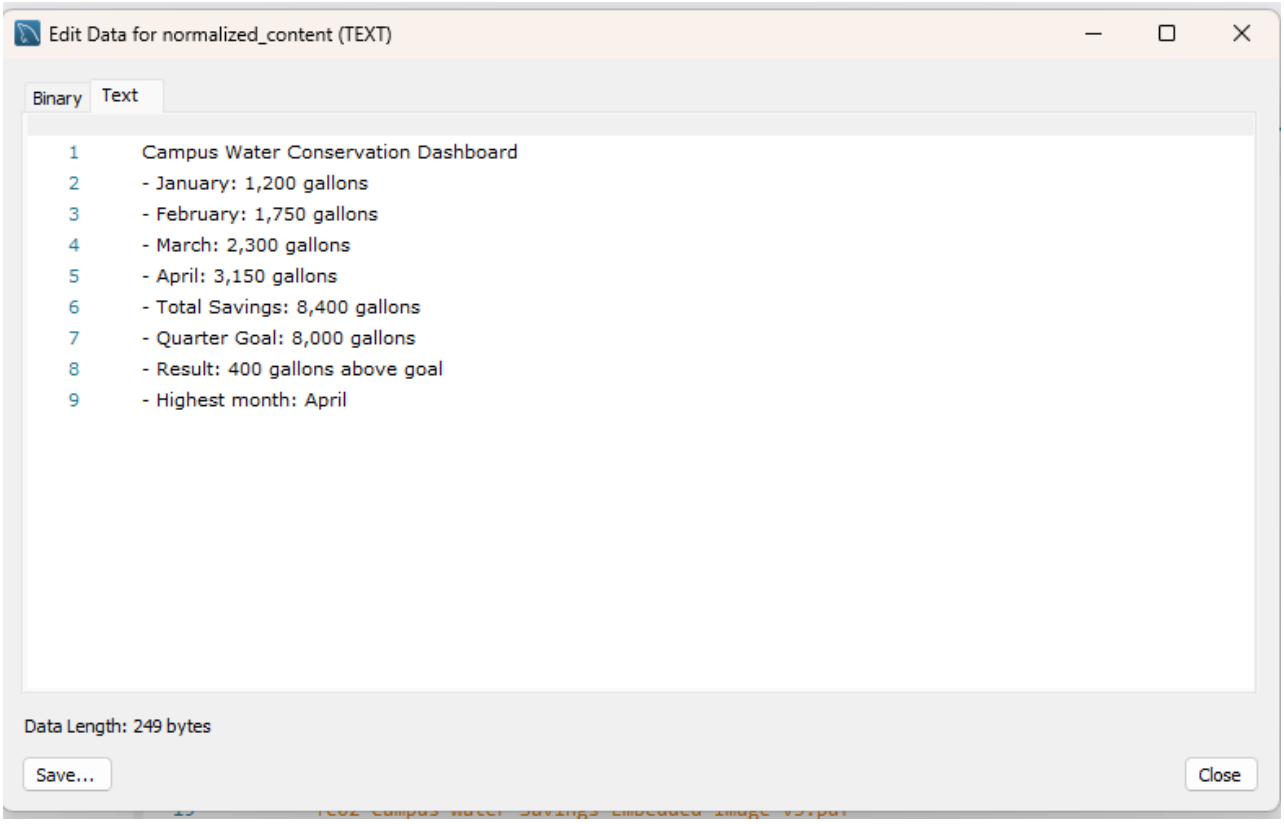


Figure 2.3 — Test Case 2 Image Chunk Description Screenshot

Chat

Monthly comparison: According to the Campus Water Conservation Dashboard, which month had the highest water savings, and how many gallons were saved?

Expected: April, 3,150 gallons.

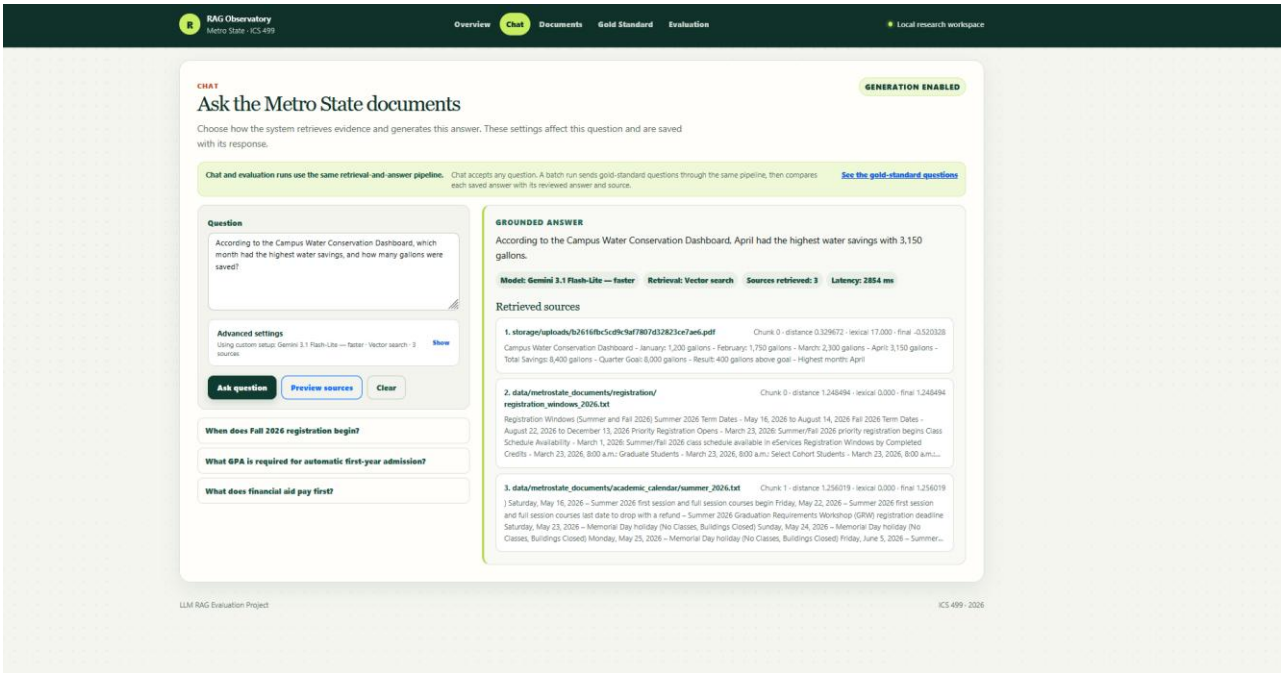


Figure 2.4 — Test Case 2 Question 1 Answer Screenshot

Goal comparison: In the Campus Water Conservation Dashboard PDF, what was the quarterly water savings goal, and by how many gallons was it exceeded?

Expected: Goal of 8,000 gallons, exceeded by 400 gallons.

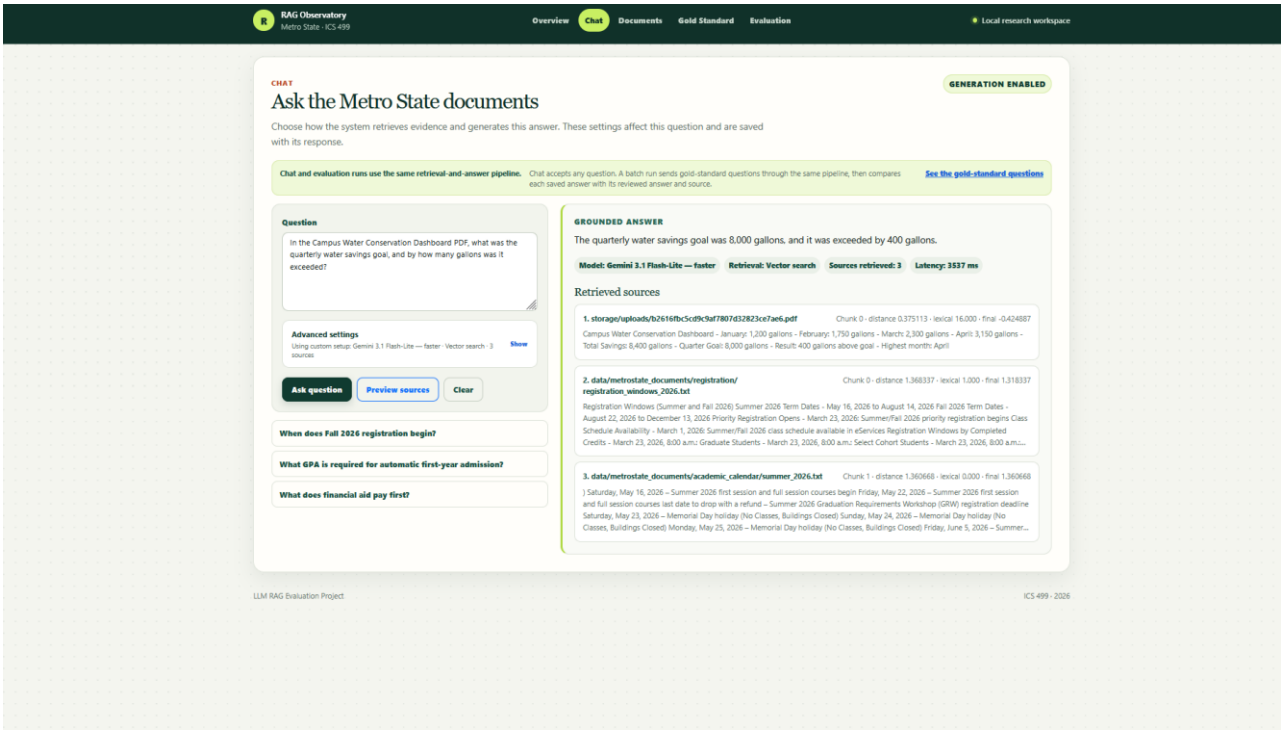


Figure 2.5 — Test Case 2 Question 2 Answer Screenshot

Numerical extraction: According to the Campus Water Conservation Dashboard, how many gallons of water were saved in February and March?

Expected: February 1,750 gallons; March 2,300 gallons.

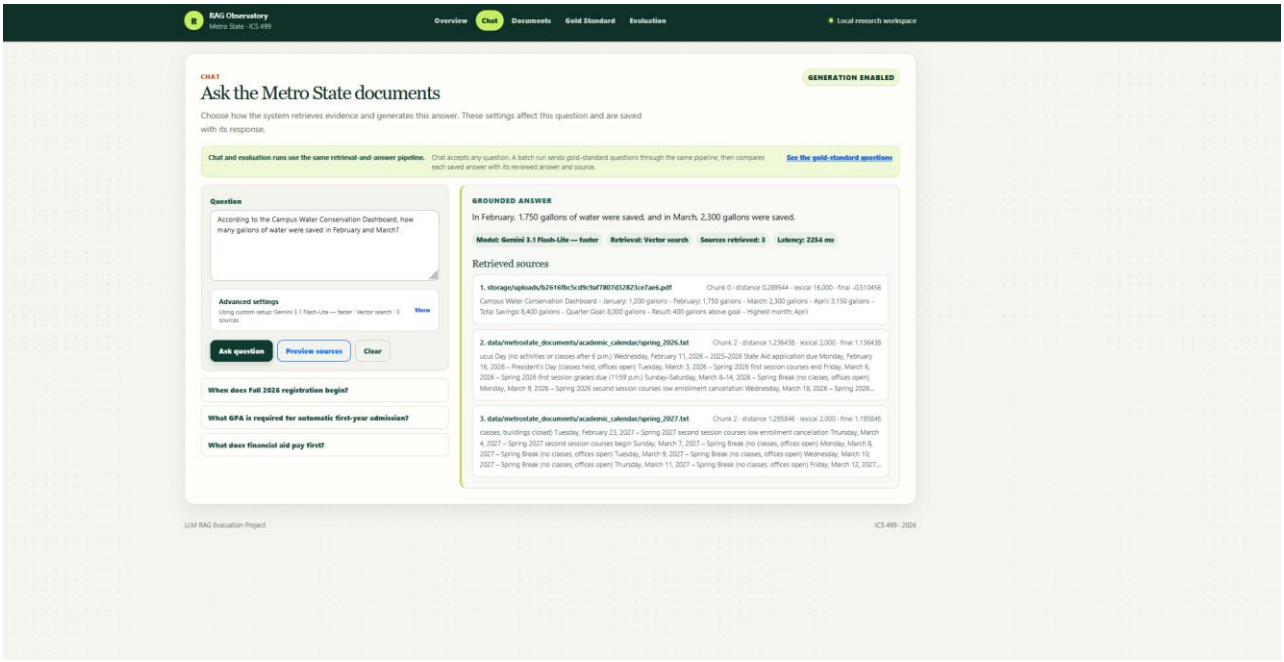


Figure 2.6 — Test Case 2 Question 3 Answer Screenshot

Assessment

PASS — supplied screenshots document ingestion and grounded answers matching the stated expected results. See individual figures and question/expected-answer notes above.

Test Case 03 — PDF containing a native table

Input fixture: TC03_Equipment_Loan_Native_Table_REGENERATED.pdf | Open / download from GitHub

Original test notes and visual evidence

Ingestion

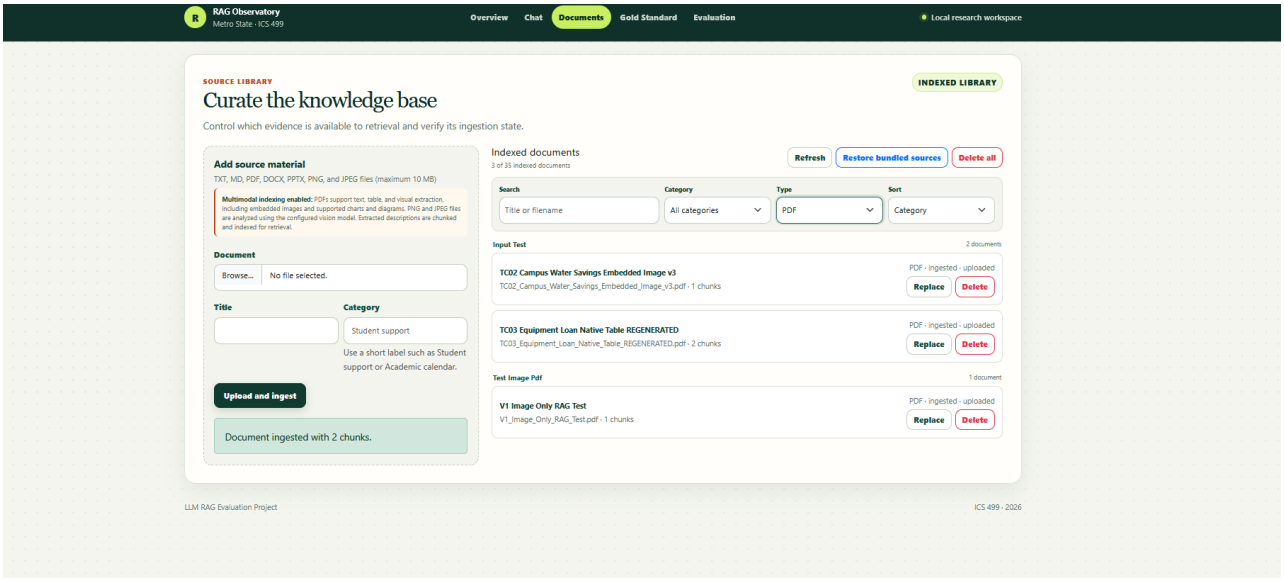


Figure 3.1 — Test Case 3 Document Ingestion Screenshot

MySQL Metadata Check

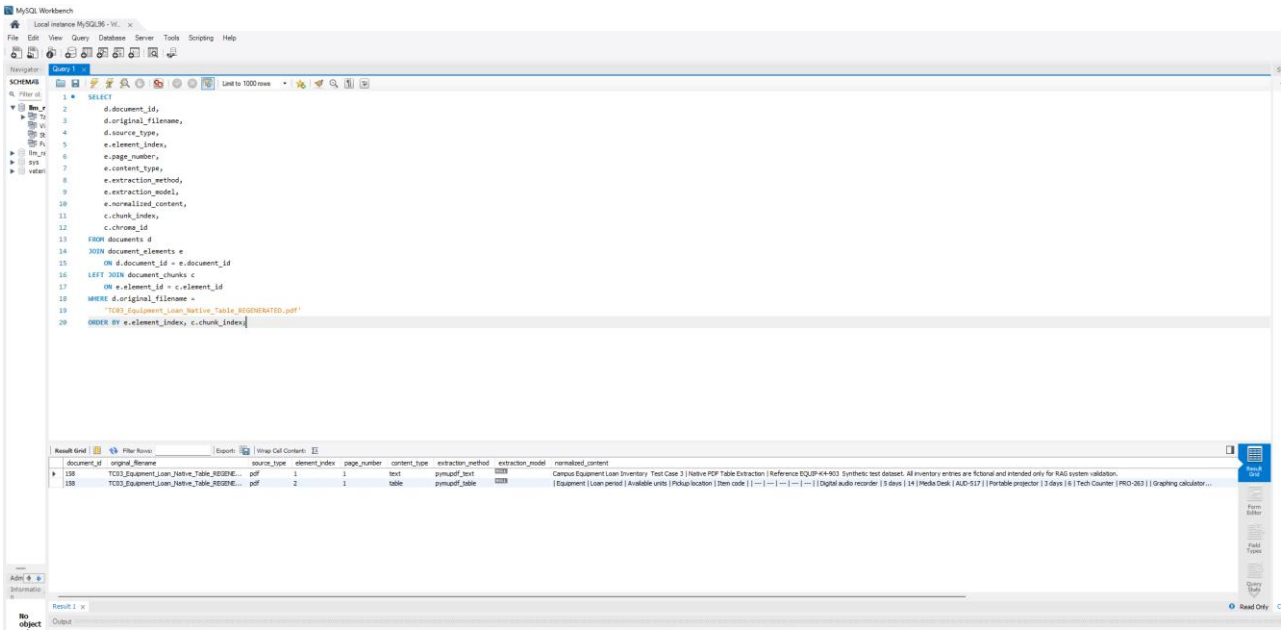


Figure 3.2 — Test Case 3 MySQL Metadata Check Screenshot

Chat

Question 1 — Equipment availability: According to the Campus Equipment Loan Inventory table, how many graphing calculators are available, and what is their loan period?

Expected: 22 available, 7-day loan period.

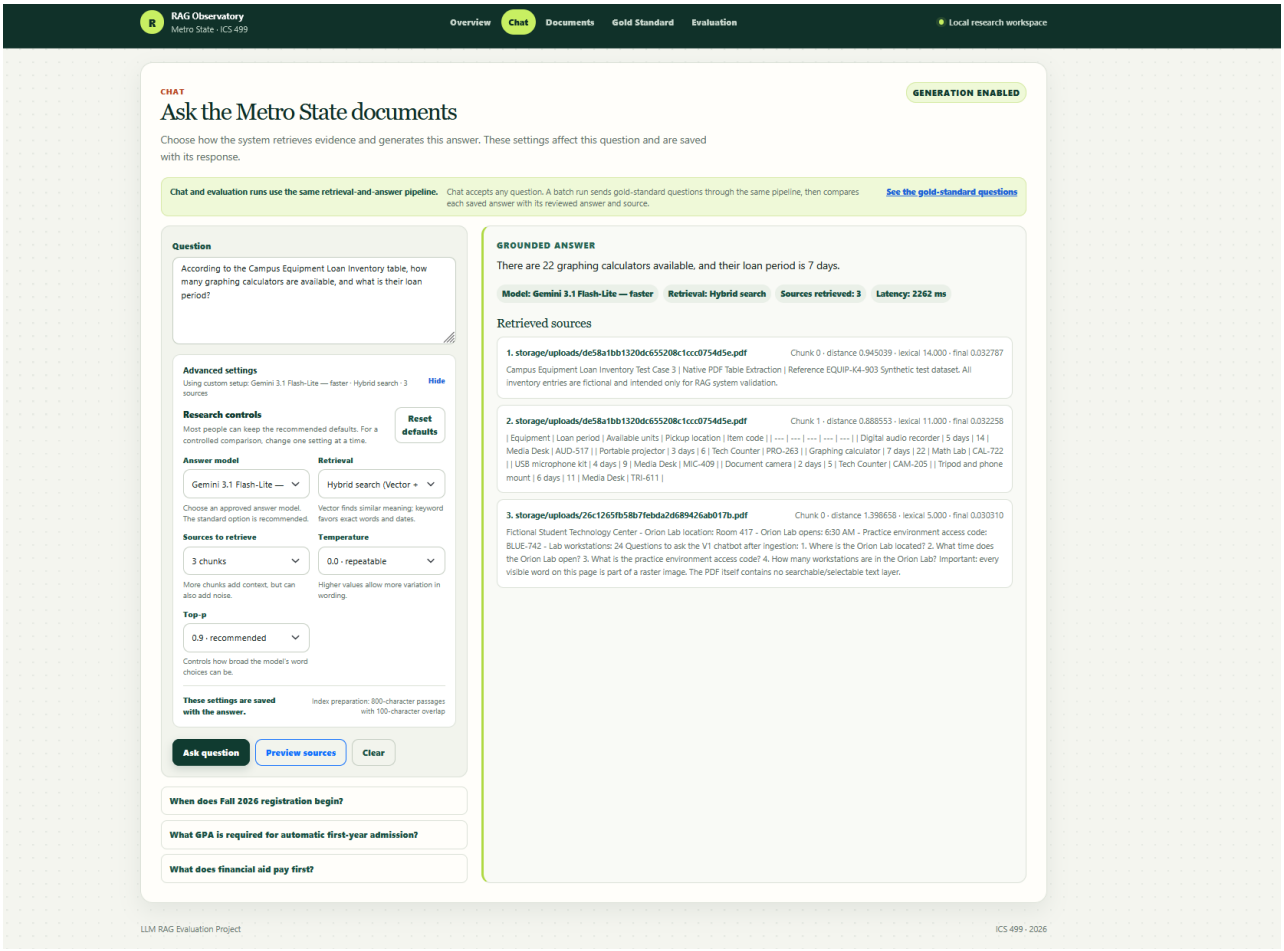


Figure 3.3 — Test Case 3 Question 1 Answer Screenshot

Question 2 — Multiple table fields: According to the Campus Equipment Loan Inventory table, what are the loan periods and available quantities for the portable projector and USB microphone kit?

Expected: Portable projector — 3 days, 6 units; USB microphone kit — 4 days, 9 units.

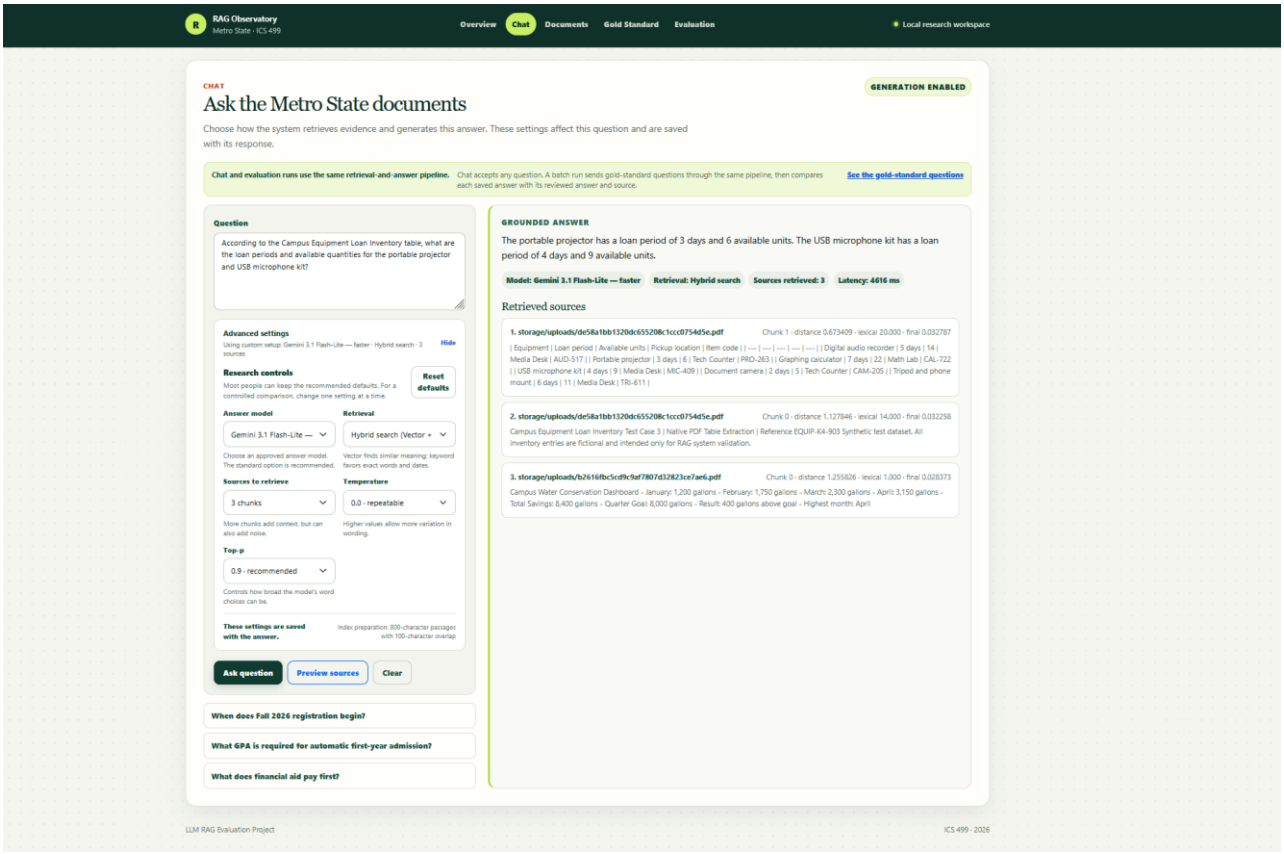


Figure 3.4 — Test Case 3 Question 2 Answer Screenshot

Assessment

PASS — supplied screenshots document ingestion and grounded answers matching the stated expected results. See individual figures and question/expected-answer notes above.

Test Case 04 — Text-only PDF

Input fixture: TC04_Bicycle_Locker_Permit_Text_Only.pdf | Open / download from GitHub

Original test notes and visual evidence

Ingestion

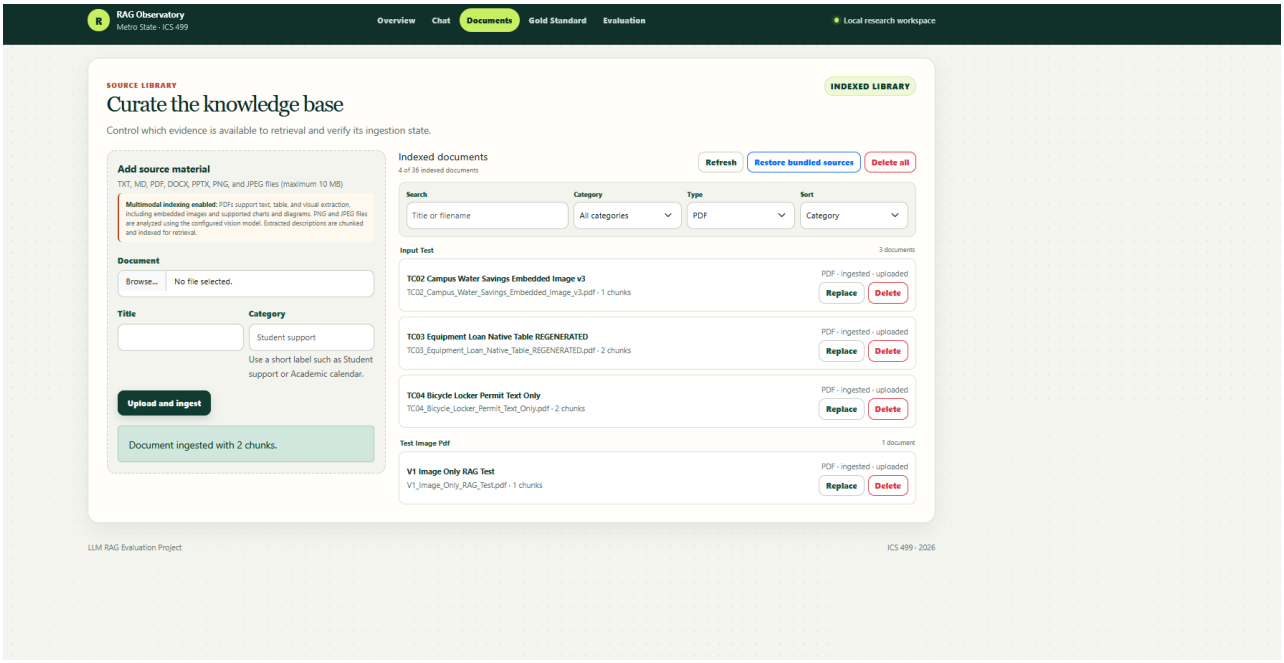


Figure 4.1 — Test Case 4 Document Ingestion Screenshot

MySQL Metadata Check

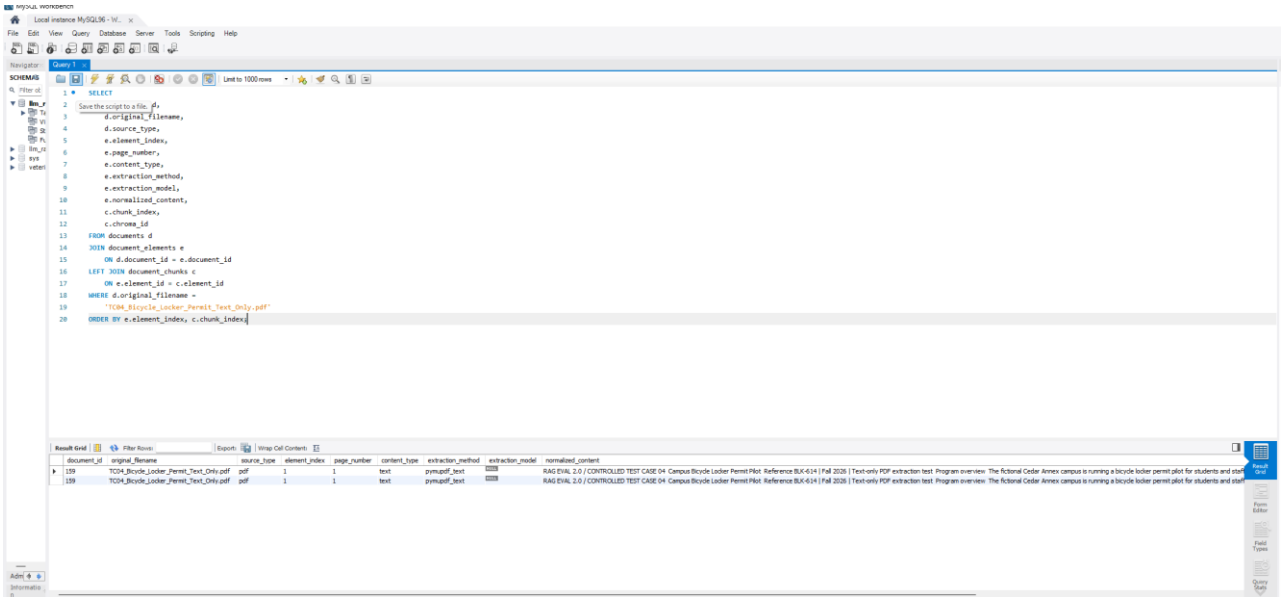


Figure 4.2 — Test Case 4 MySQL Metadata Check Screenshot



Figure 4.3 — Test Case 4 Text Chunk Description Screenshot

Normalized_content (description of chunk) *duplicate text chunk – Investigate further

Chat

Question 1 — Application dates: According to the Campus Bicycle Locker Permit Pilot document, when do applications open and close?

Expected: October 26, 2026, through November 6, 2026.

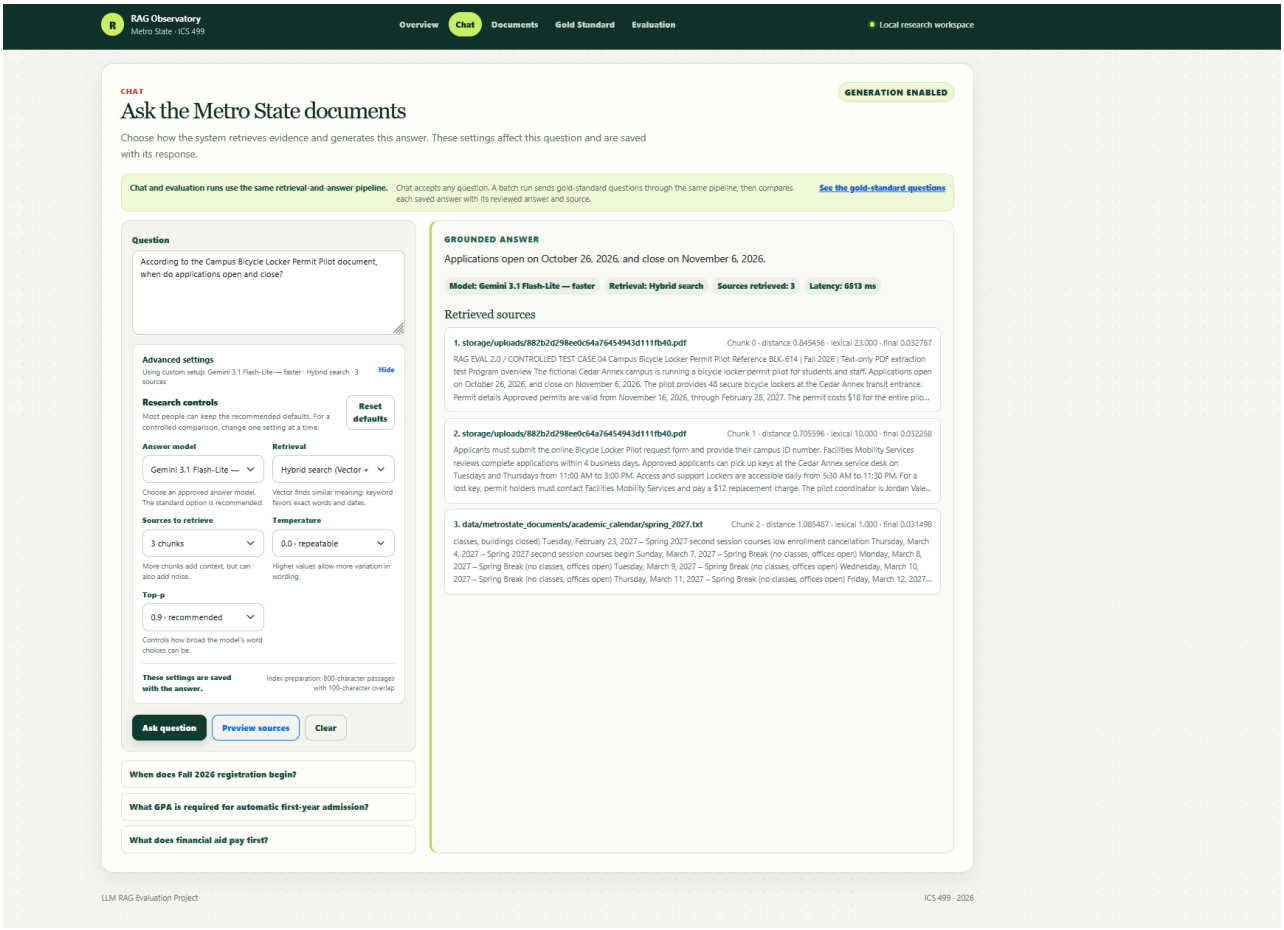


Figure 4.4 — Test Case 4 Question 1 Answer Screenshot

Question 2 — Permit cost and deposit: According to the Campus Bicycle Locker Permit Pilot document, how much does the permit cost, what is the refundable key deposit, and when is the deposit returned?

Expected: $18 permit, $25 refundable key deposit, returned within 10 business days after the key is surrendered in good condition.

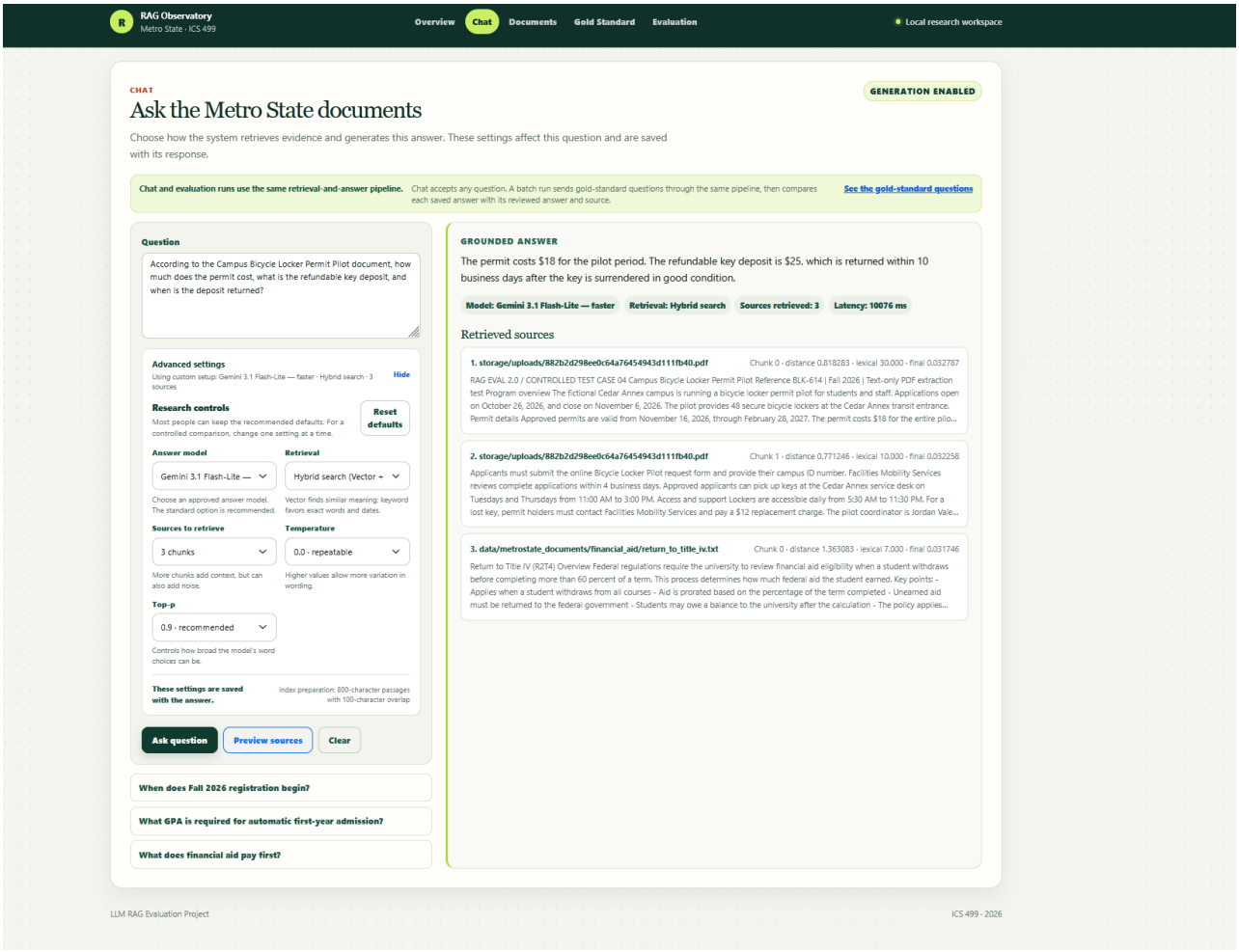


Figure 4.5 — Test Case 4 Question 2 Answer Screenshot

Question 3 — Application review and key pickup: According to the Campus Bicycle Locker Permit Pilot document, how long does Facilities Mobility Services take to review applications, and when and where can approved applicants pick up their keys?

Expected: Within 4 business days; Tuesdays and Thursdays, 11:00 AM–3:00 PM, at the Cedar Annex service desk.

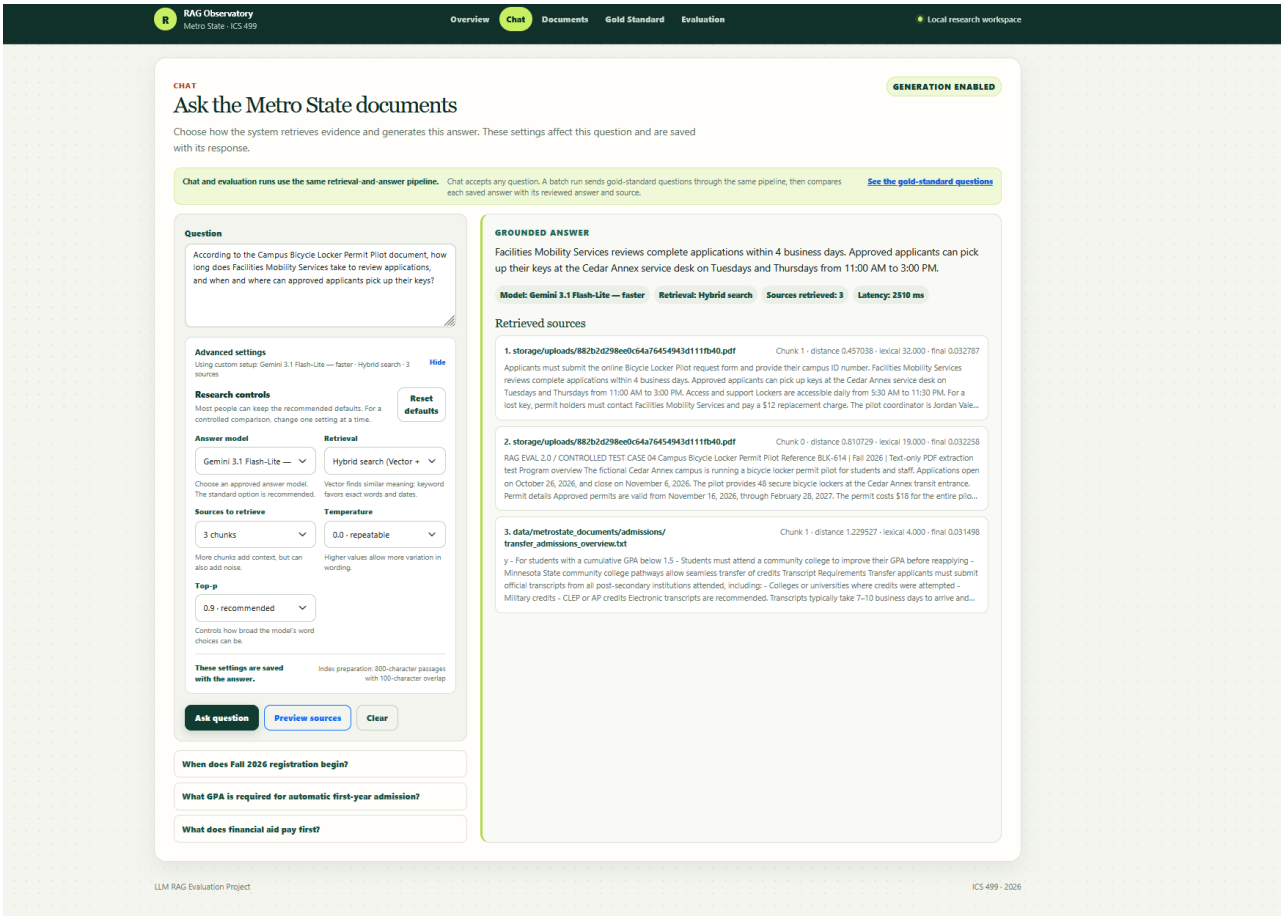


Figure 4.6 — Test Case 4 Question 3 Answer Screenshot

Assessment

PASS — supplied screenshots document ingestion and grounded answers matching the stated expected results. See individual figures and question/expected-answer notes above.

Test Case 05 — PDF with text, table, and image

Input fixture: TC05_Campus_Greenhouse_Orientation_Mixed_Content.pdf | Open / download from GitHub

Original test notes and visual evidence

Ingestion

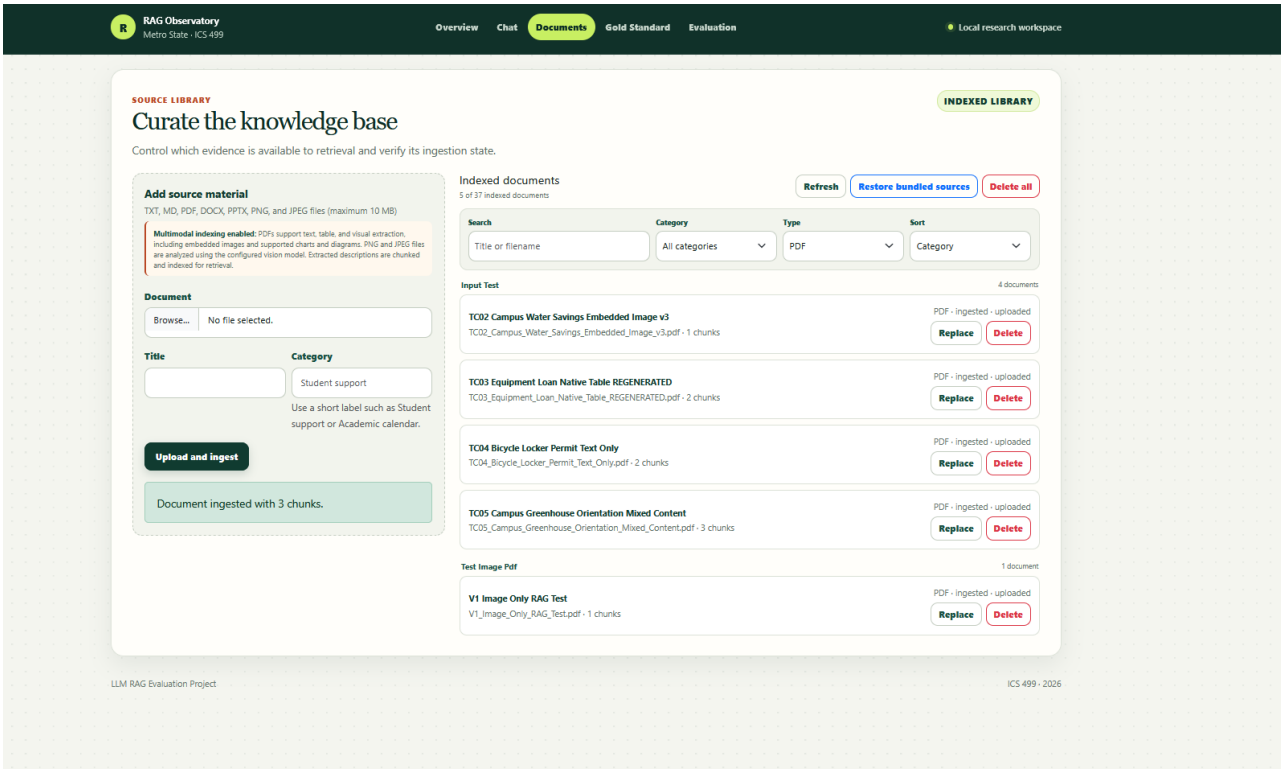


Figure 5.1 — Test Case 5 Document Ingestion Screenshot

MySQL Metadata Check

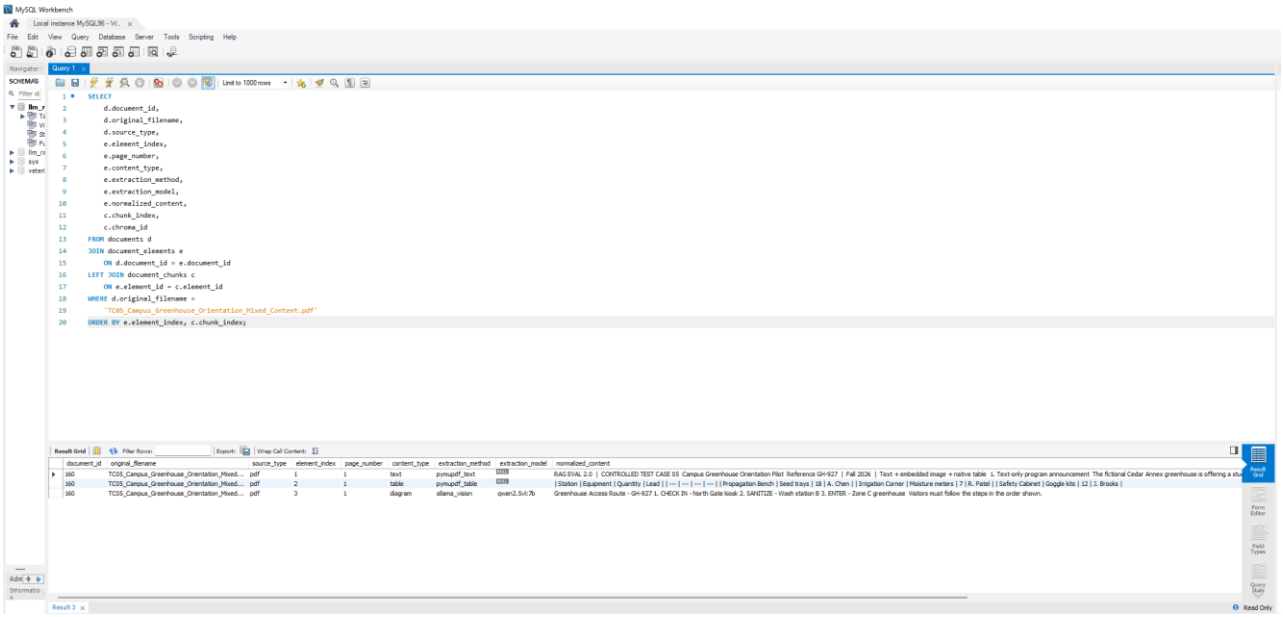


Figure 5.2 — Test Case 5 MySQL Metadata Check Screenshot

Question 1 — Text According to the Campus Greenhouse Orientation Pilot document (GH-927), what is the greenhouse orientation program and who is it intended for?

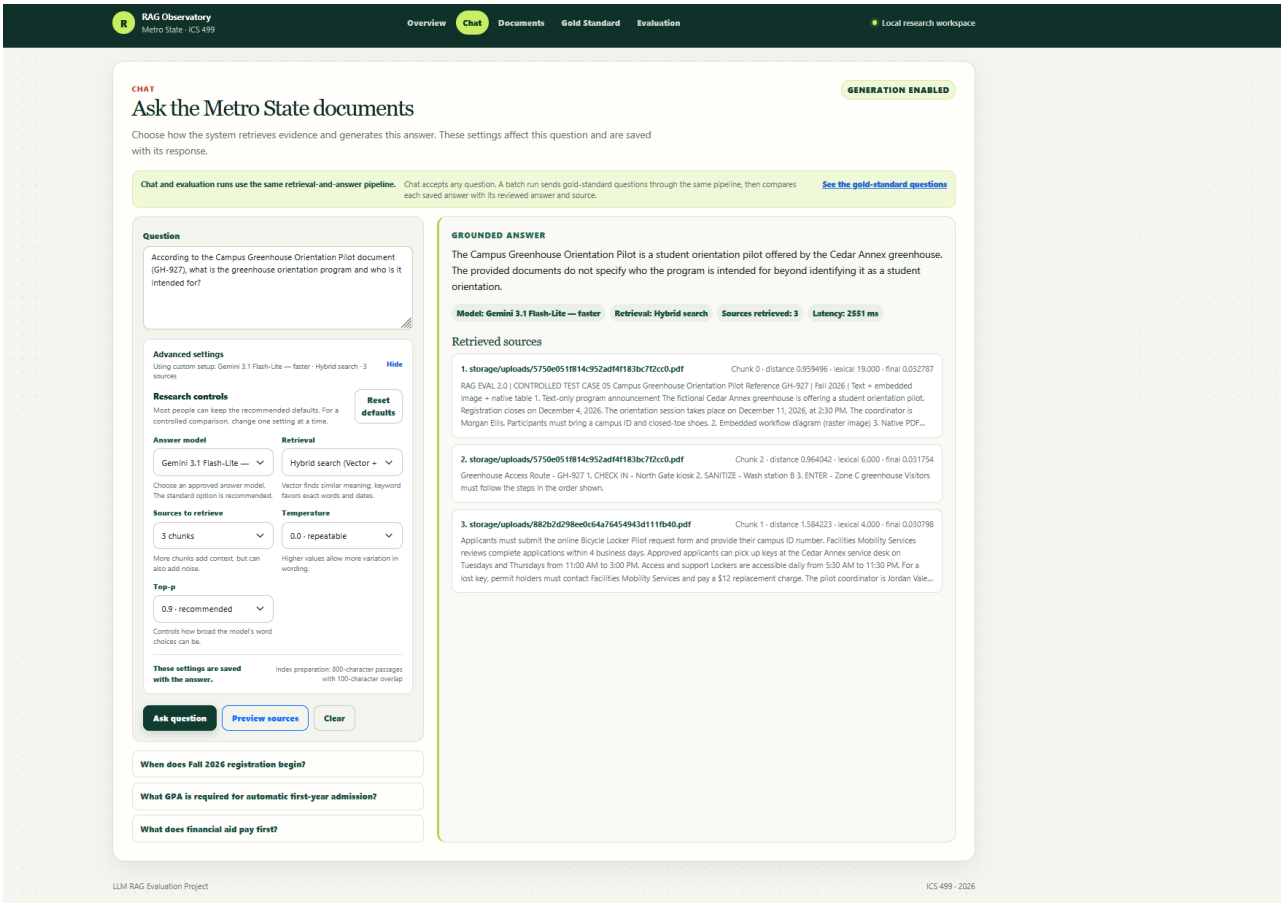


Figure 5.3 — Test Case 5 Question 1 Text Retrieval Answer Screenshot

Expected Answer: Identify the greenhouse orientation pilot

Question 2 — Table Retrieval: According to the Campus Greenhouse Orientation equipment table, how many seed trays are available, and who is responsible for them?

Expected Answer: 18 seed trays, A. Chen

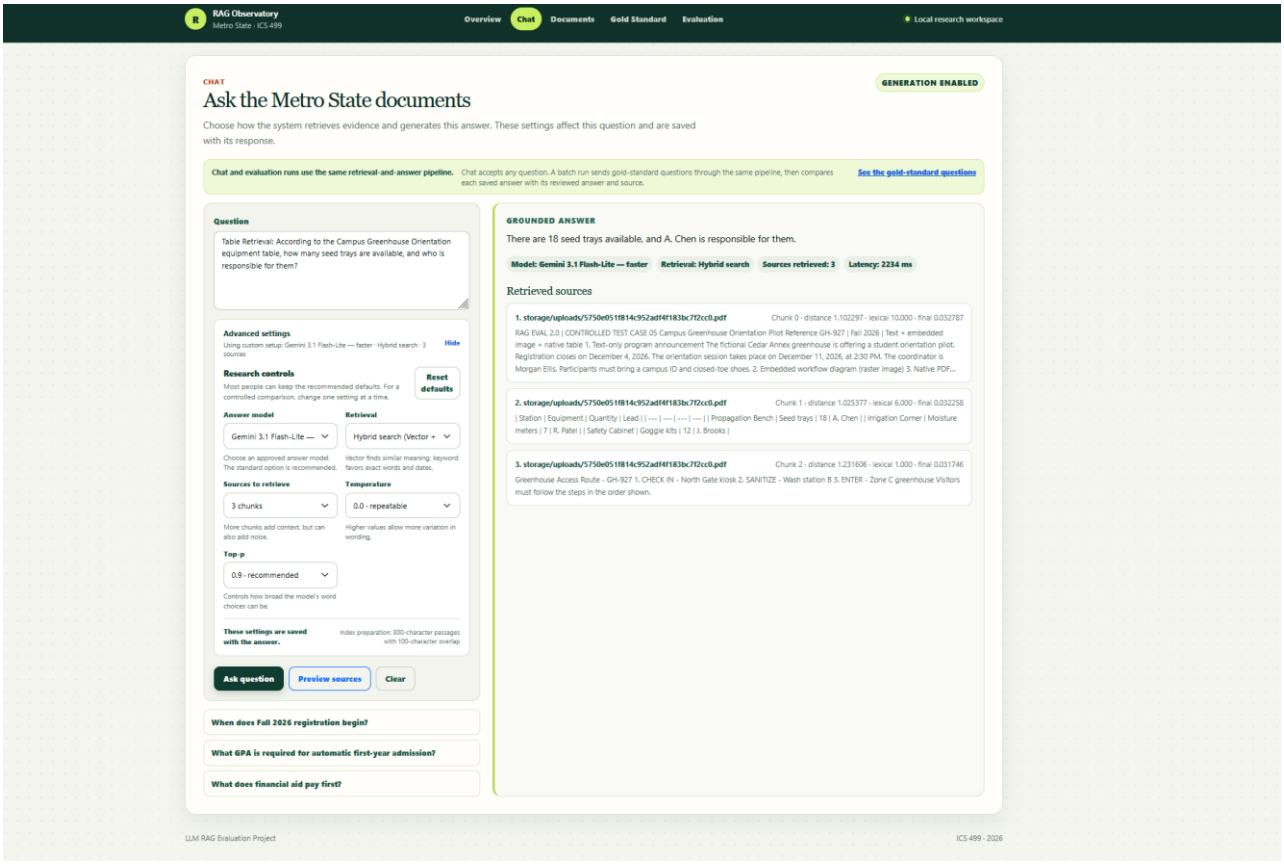


Figure 5.4 — Test Case 5 Question 2 Table Retrieval Answer Screenshot

Question 3 — Diagram Retrieval: According to the Greenhouse Access Route diagram GH-927, what are the three steps visitors must follow to enter the greenhouse?

Expected Answer: Check In → Sanitize → Enter

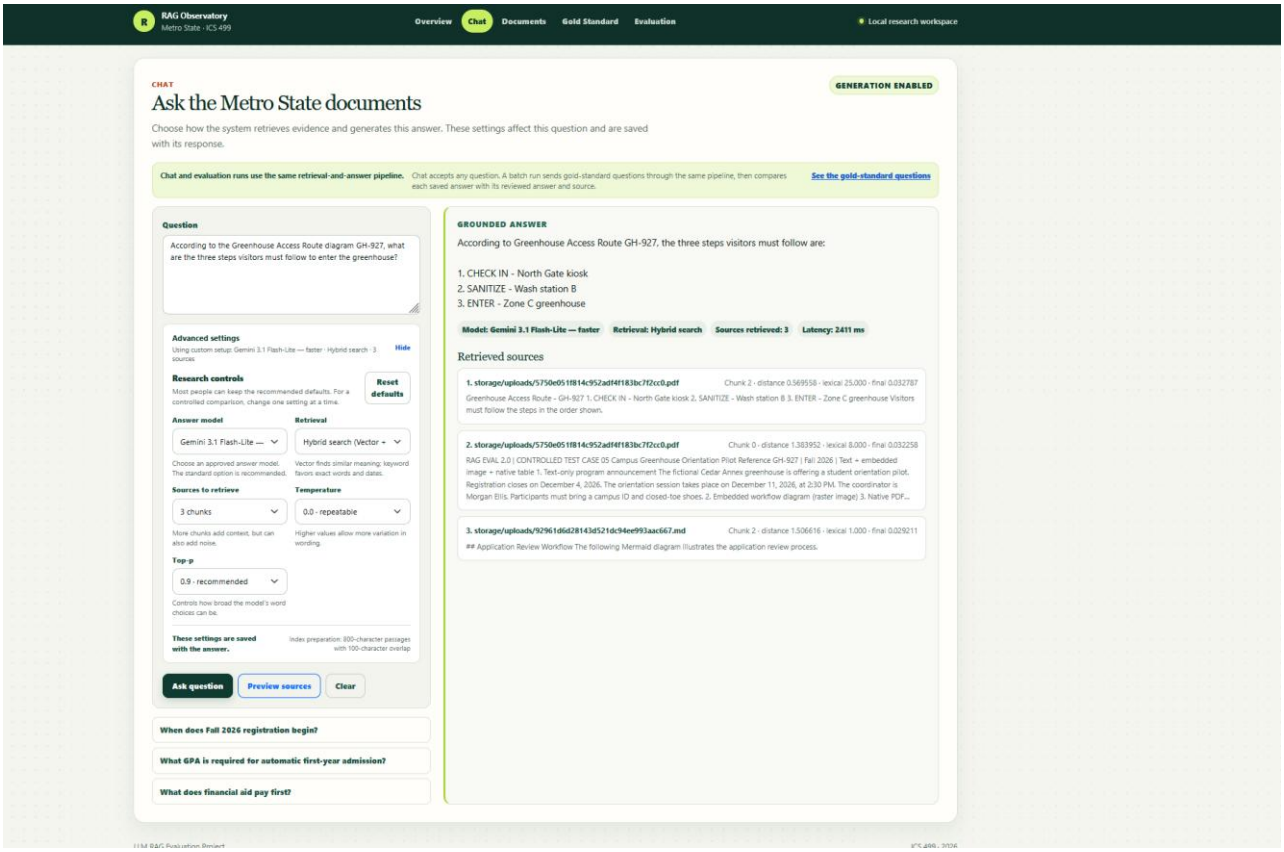


Figure 5.5 — Test Case 5 Question 3 Diagram Retrieval Answer Screenshot

Assessment

PASS — supplied screenshots document ingestion and grounded answers matching the stated expected results. See individual figures and question/expected-answer notes above.

Test Case 06 — PDF with two text sections, two tables, and two images

Input fixture: TC06_Community_Repair_Lab_Mixed_Content.pdf | Open / download from GitHub

Retrieval improvement implemented during testing

The original top-three retrieval results did not consistently include the intake table or collection-route diagram. Table chunks were enriched with short natural-language field and row descriptions while preserving the original table in the stored element and chunk. This made the table more searchable and improved its position in the candidate ranking (observed from approximately rank 7–8 to rank 4).

A lightweight rule-based content-aware reranker was then introduced. It examines terms in the question (such as "table," "quantities," "diagram," and "steps") and overlap with candidate evidence to promote relevant content types. It operates after vector, keyword, or hybrid retrieval, with an internal opt-out for baseline comparison. In the final browser checks, the intake table appeared at source 3 of 3 and the collection-route diagram appeared at source 1 of 3. Both generated answers matched the expected results.

Known observation: the collection-route diagram was described through two extraction paths, producing overlapping evidence. This duplication remains a potential future optimization.

Original test notes and visual evidence

Ingestion

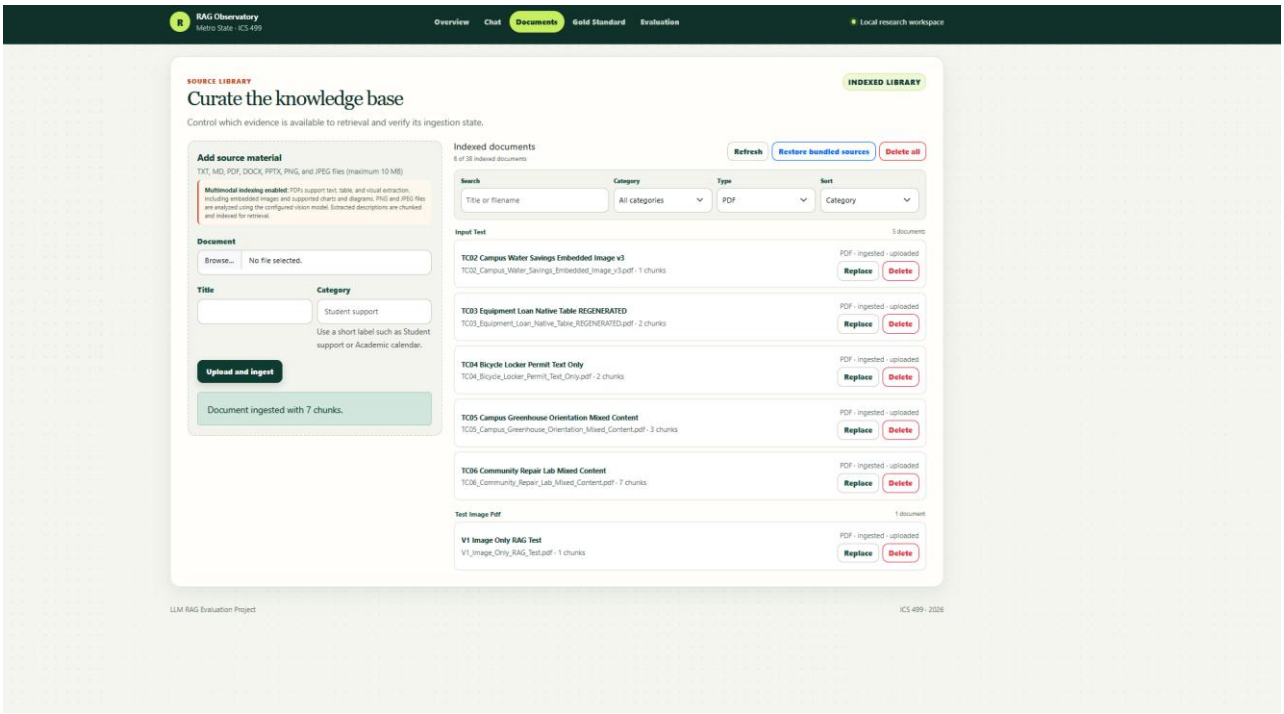


Figure 6.1 — Test Case 6 Document Ingestion Screenshot

MySQL Metadata Check

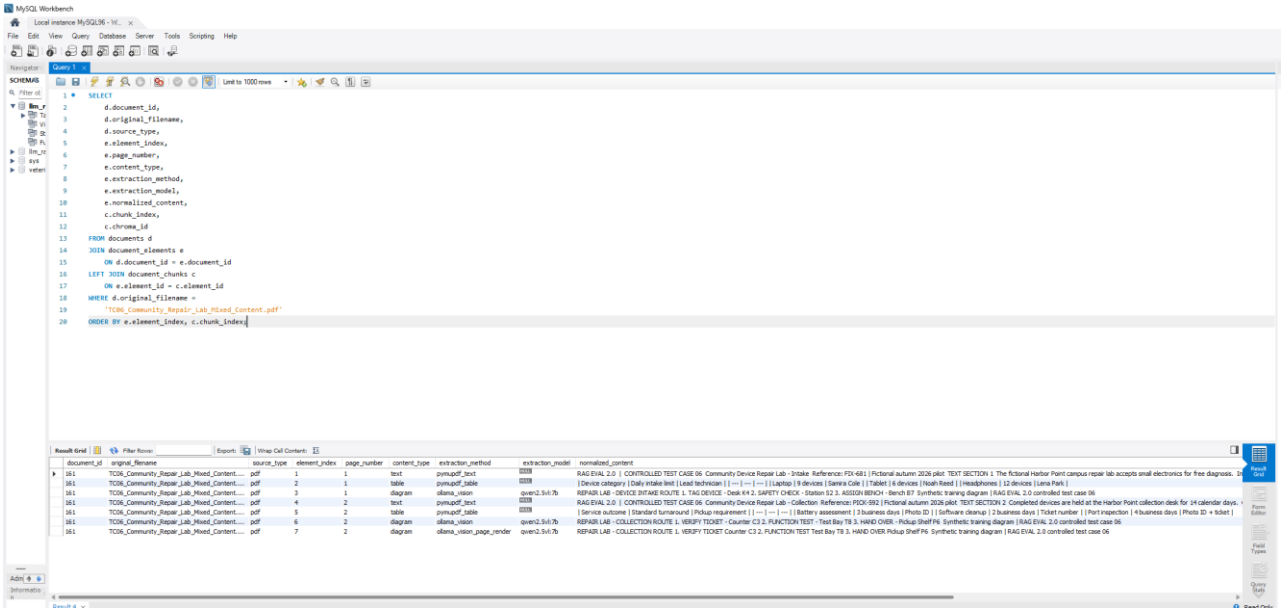


Figure 6.2 — Test Case 6 MySQL Metadata Check Screenshot

Duplicate image detected (Element 7)

Instead of extracting an embedded image, it rendered part of the PDF page as an image and sent that rendering to the same vision model.

Looking at the database results, both elements describe the same Collection Route:

Verify Ticket → Function Test → Hand Over

This strongly suggests the second diagram was processed twice through two different extraction methods. - Improve

Chat

Question 1 — Text Extraction: According to the Community Device Repair Lab document (FIX-681), what types of devices does the repair lab accept, and what service does it provide?

Expected Answer

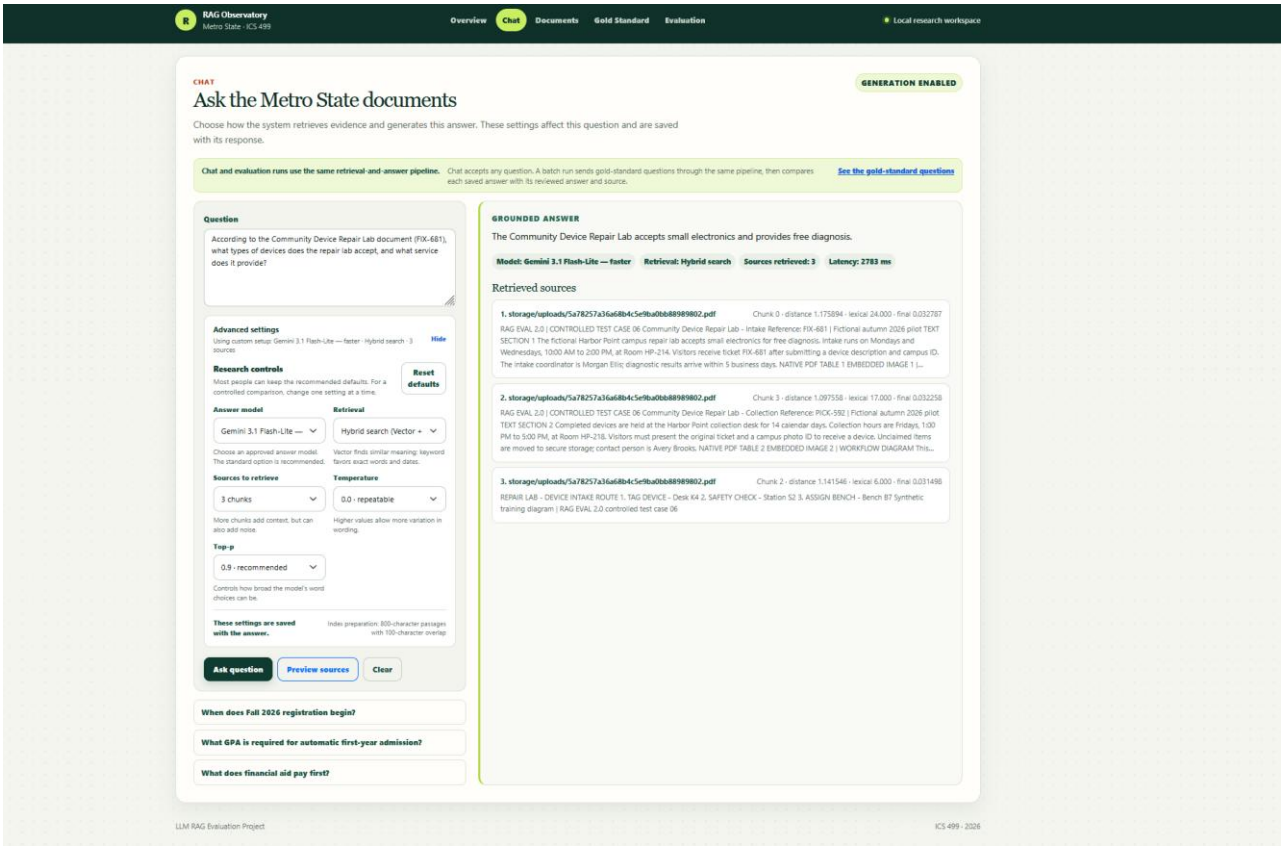


Figure 6.3 — Test Case 6 Question 1 Text Retrieval Answer Screenshot

Question 2 — According to the Community Device Repair Lab intake table (FIX-681), what are the intake quantities and assigned leads for the device categories listed?

Expected Answer

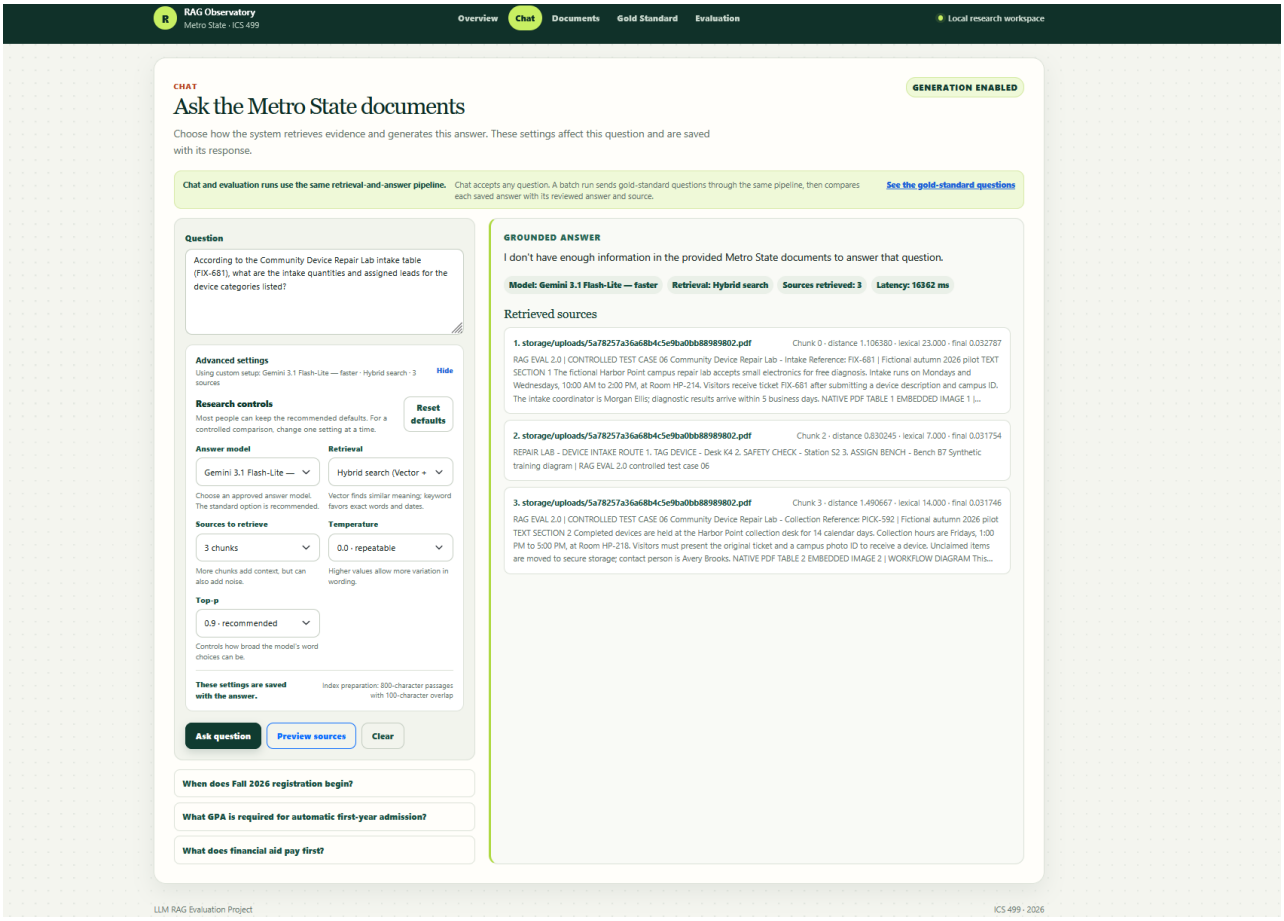


Figure 6.4 — Test Case 6 Question 2 Table Retrieval Answer Screenshot

After applying natural language enrichment to table chunks alongside the table structure chunking ranking of the table moved from 8 -> 4 then after applying a rule based reranker the retrieval of both tables and diagrams significantly improved.

For this test the correct table chunk entered top 3 retrieved sources.

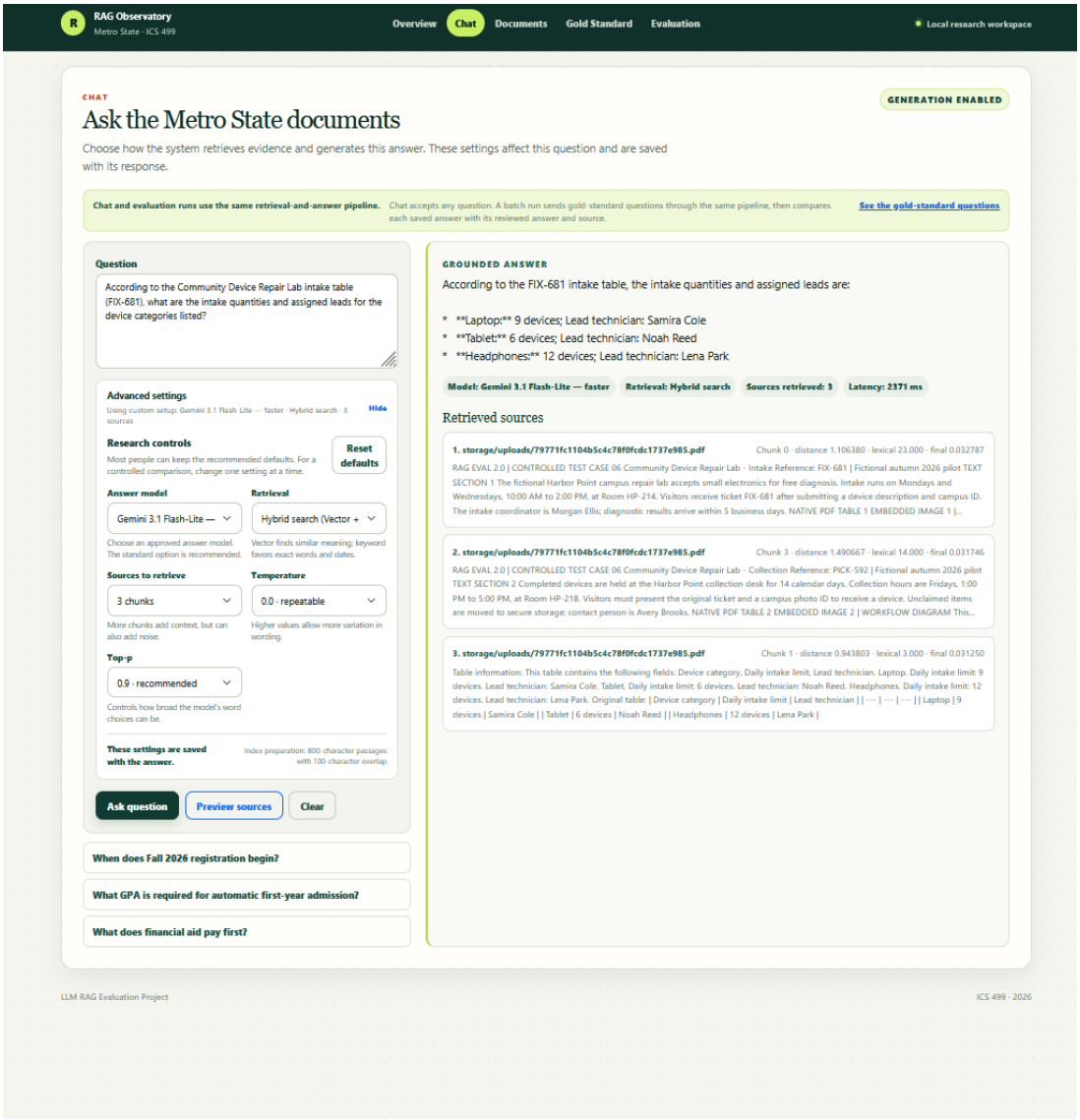


Figure 6.5 — Test Case 6 Table Retrieval Enhancement Evidence Screenshot 1

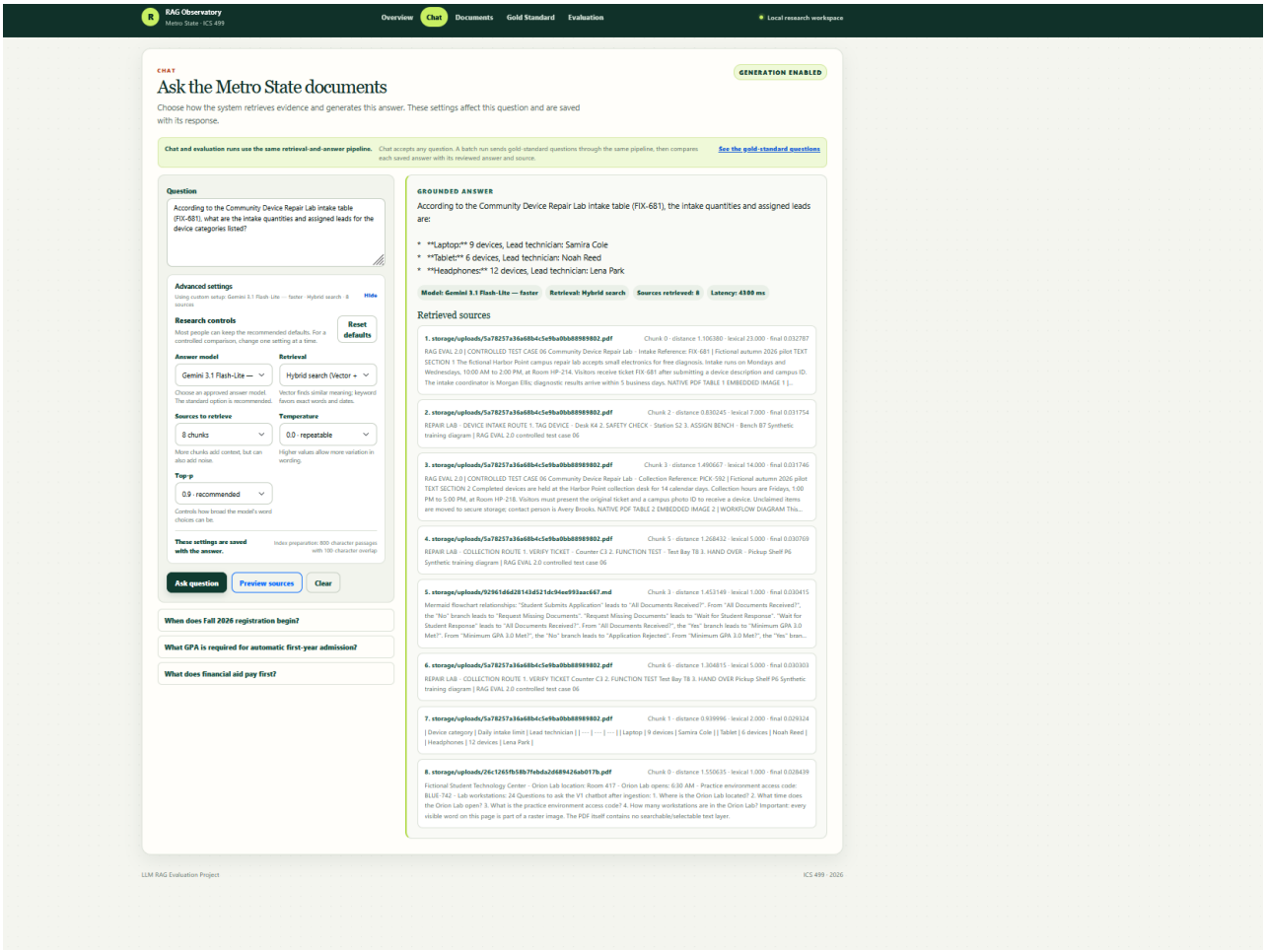


Figure 6.6 — Test Case 6 Table Retrieval Enhancement Evidence Screenshot 2

Question 3 — According to the Repair Lab Collection Route diagram (FIX-681), what are the three steps visitors must follow to collect their repaired devices, in the correct order?

Expected answer

Verify Ticket — Counter C3

Function Test — Test Bay T8

Hand Over — Pickup Shelf P6

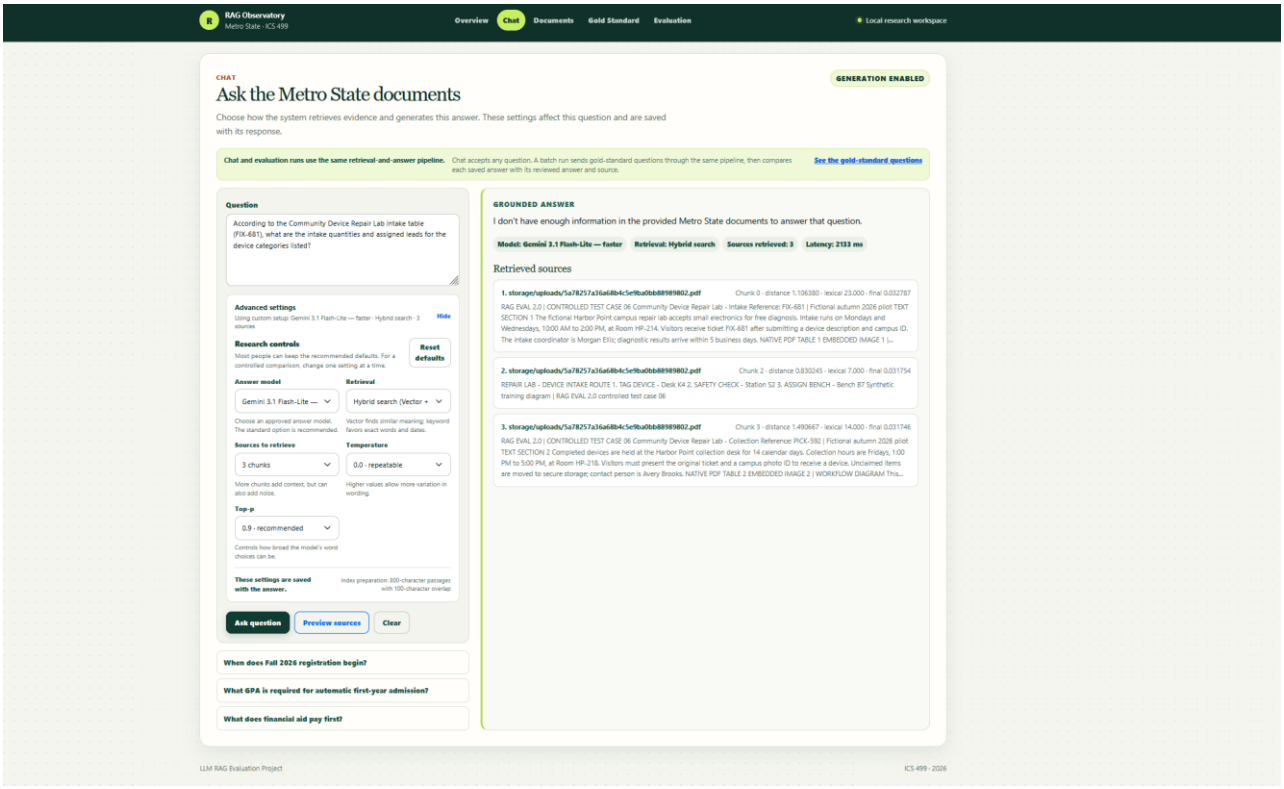


Figure 6.7 — Test Case 6 Question 3 Diagram Retrieval Answer Screenshot

For diagrams the reranker helped improve the correct retrieval of diagram and for this specific test case improved the correct diagram resource to number 1 ranked source.

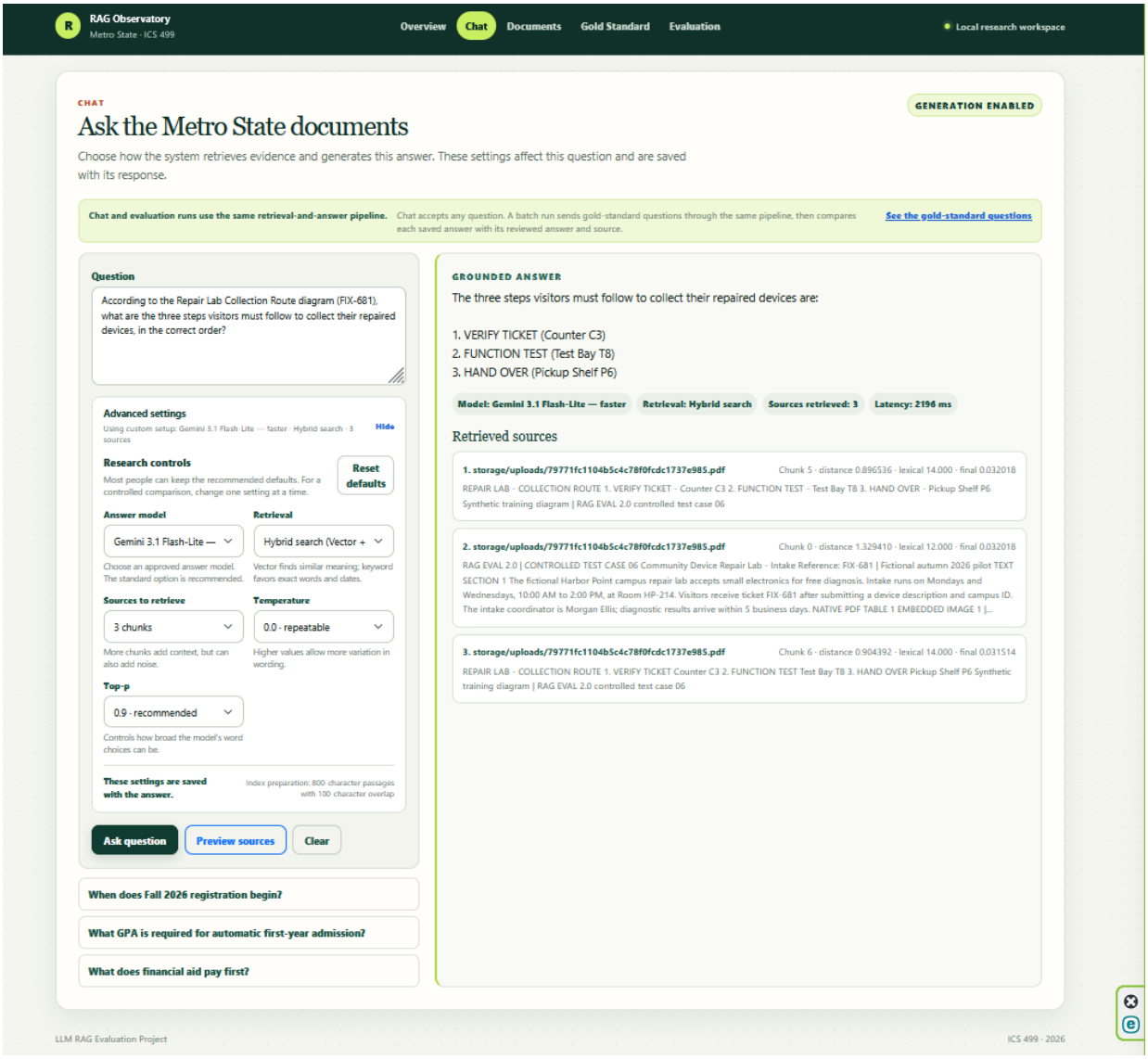


Figure 6.8 — Test Case 6 Diagram Reranking Evidence Screenshot 1

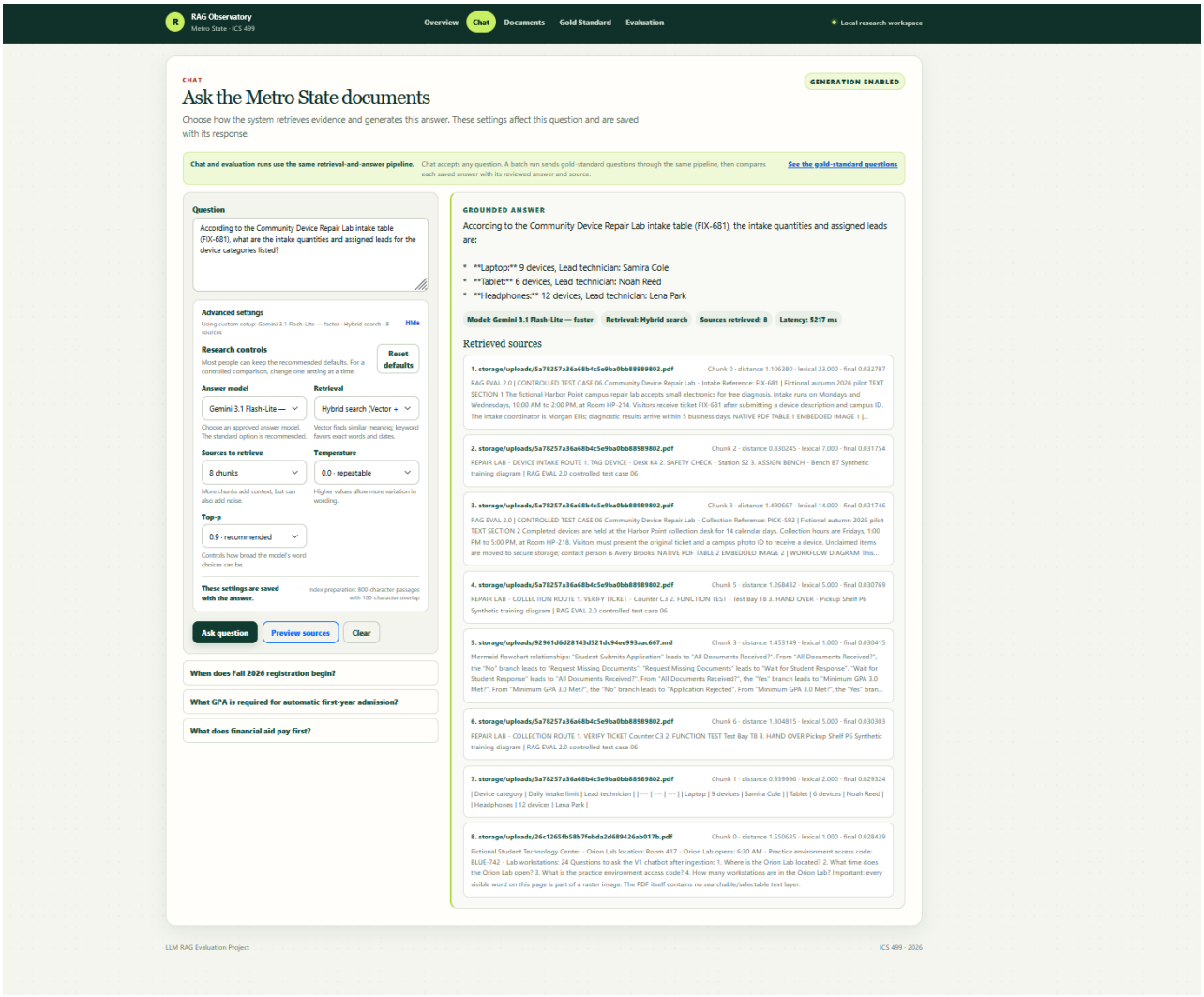


Figure 6.9 — Test Case 6 Diagram Reranking Evidence Screenshot 2

Assessment

PASS — final browser tests confirm the expected intake quantities and assigned leads, and the correct three-step collection route, with Hybrid RRF and Top K = 3. A duplicate visual-description observation remains open.

Test Case 07 — Formatted Markdown

Input fixture: TC07_markdown_formatted_test.md | Open / download from GitHub

Formatted Markdown support

The Markdown ingestion path preserves headings, formatted prose, and structured source content as searchable text chunks. This allows normal text questions to retrieve workshop details while retaining a grounded refusal for information absent from the document (such as an unspecified parking fee).

Original test notes and visual evidence

Ingestion

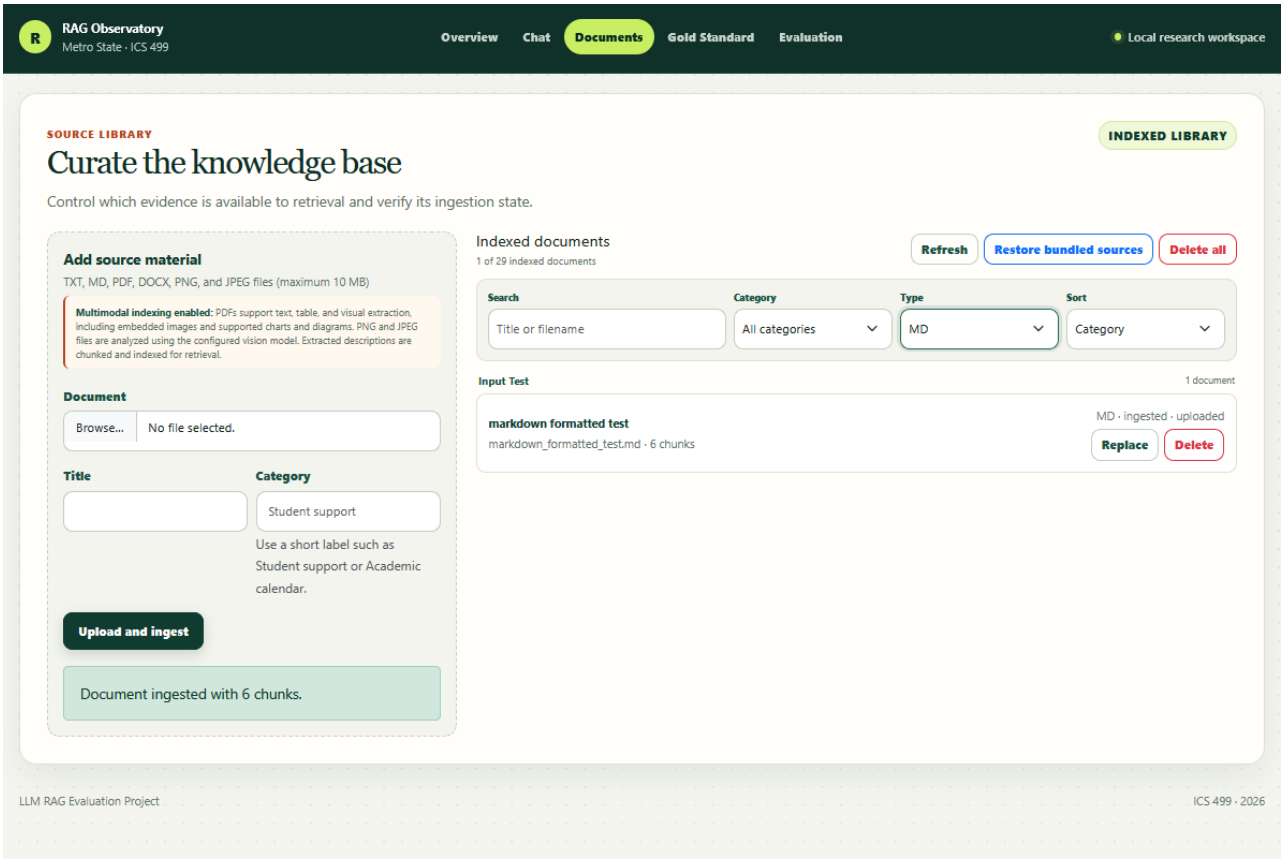


Figure 7.1 — Test Case 7 Document Ingestion Screenshot

Chat test

According to the Metro State Student Technology Workshop document, on what date will the workshop take place, and in which room will it be held?

EXPECTED ANSWER

November 12, 2026, in Room 305 at the Saint Paul Campus.

ACTUAL ANSWER

"The workshop will take place on November 12, 2026, in Room 305."

According to the Metro State Student Technology Workshop document, who is teaching the Data Analytics session, and what time does it begin?

EXPECTED ANSWER

Prof. Marcus Chen, at 11:00 AM.

ACTUAL ANSWER

"The Data Analytics session is taught by Prof. Marcus Chen and begins at 11:00 AM."

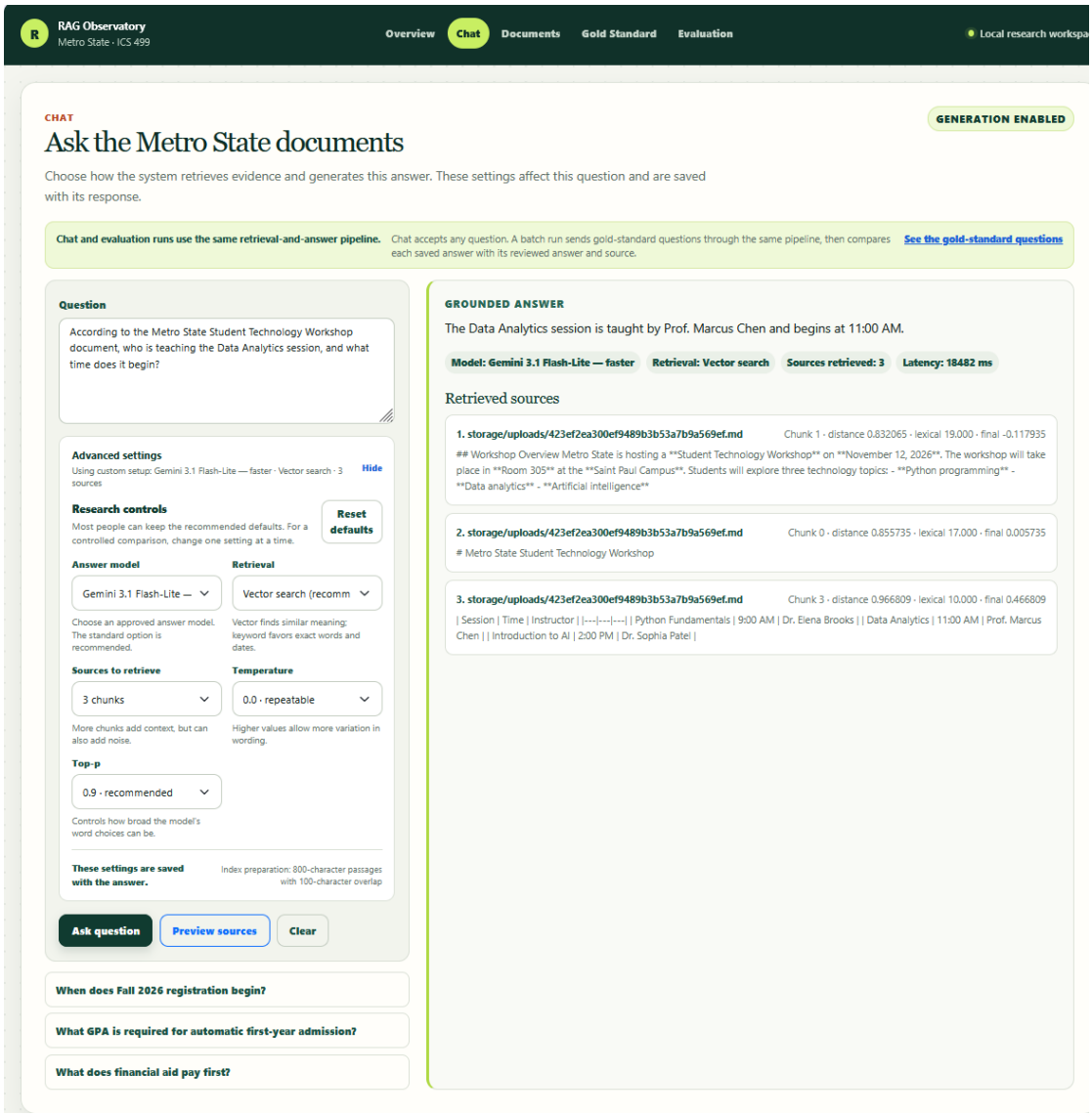


Figure 7.2 — Test Case 7 Question 1 Chat Evidence Screenshot 1

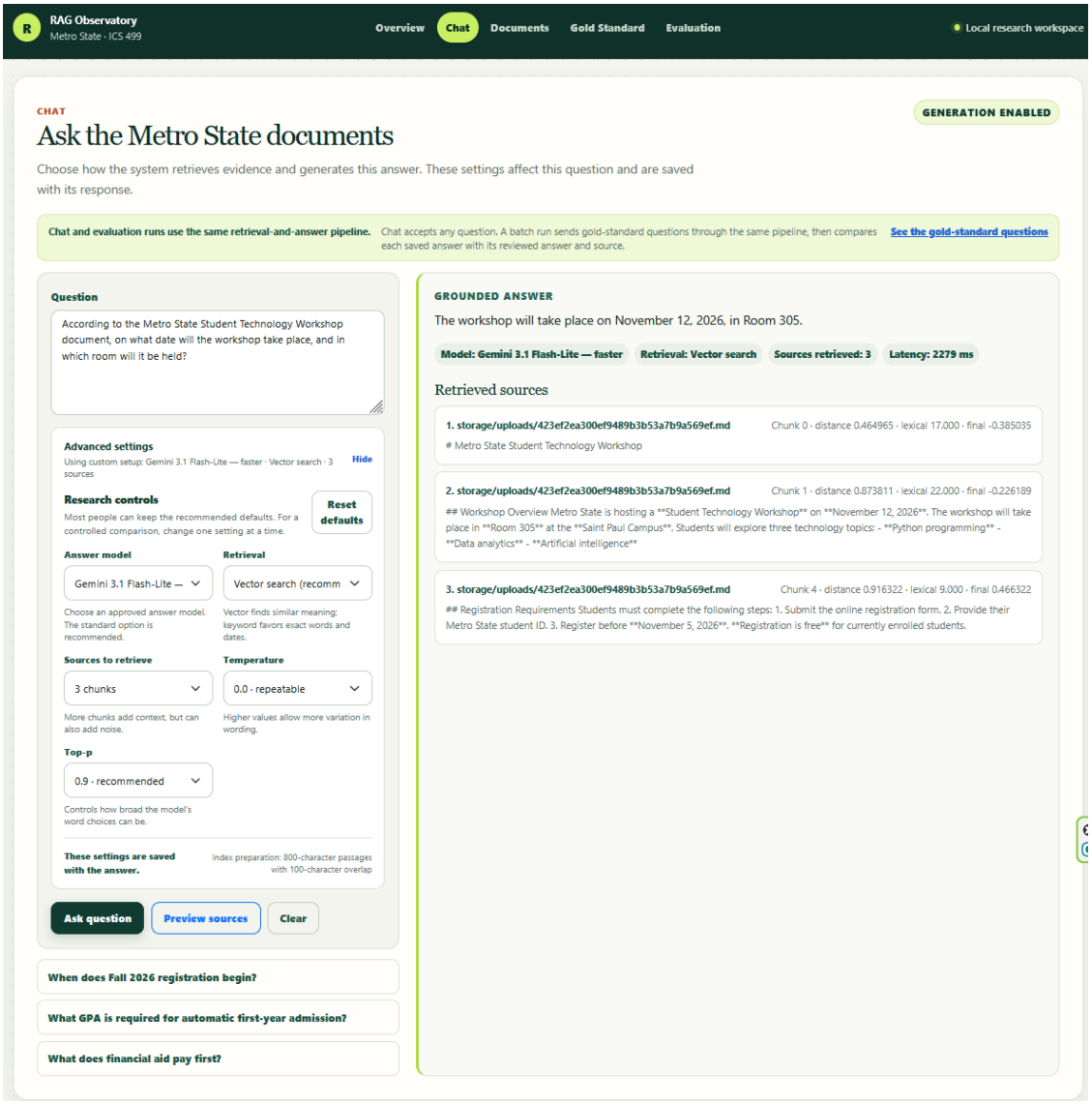


Figure 7.3 — Test Case 7 Question 1 Chat Evidence Screenshot 2

According to the Metro State Student Technology Workshop document, what is the registration deadline, and how much does registration cost for currently enrolled students?

EXPECTED ANSWER

Registration must be completed before November 5, 2026, and registration is free for currently enrolled students.

ACTUAL ANSWER

"The registration deadline for the Student Technology Workshop is November 5, 2026, and registration is free for currently enrolled students."

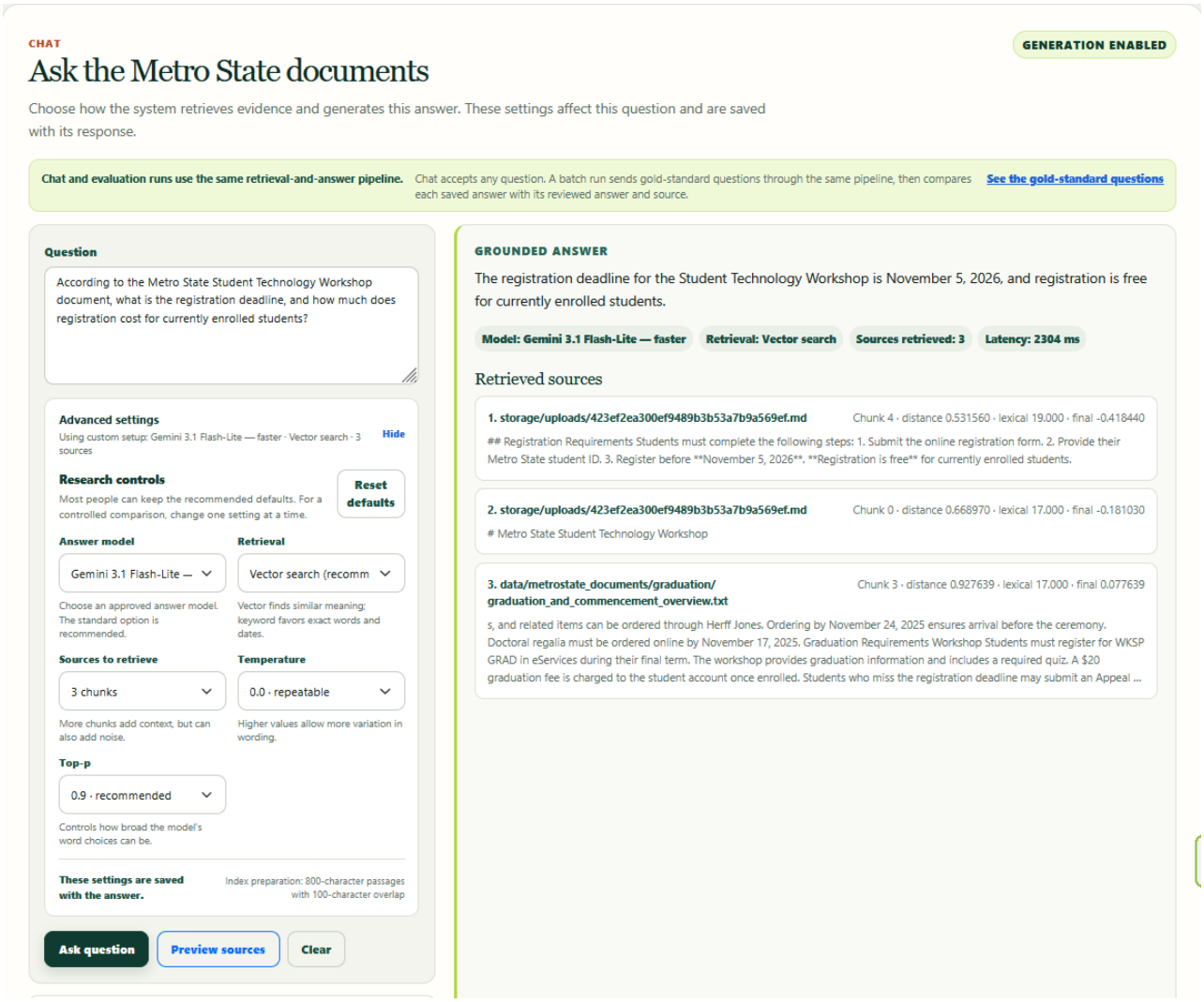


Figure 7.4 — Test Case 7 Question 2 Answer Screenshot

Unanswerable question: According to the Metro State Student Technology Workshop document, what is the exact parking fee for students attending the workshop on November 12, 2026?

EXPECTED BEHAVIOR

Acknowledge that the workshop document does not specify a parking fee. Do not invent a dollar amount.

ACTUAL ANSWER

"I don't have enough information in the provided Metro State documents to answer that question."

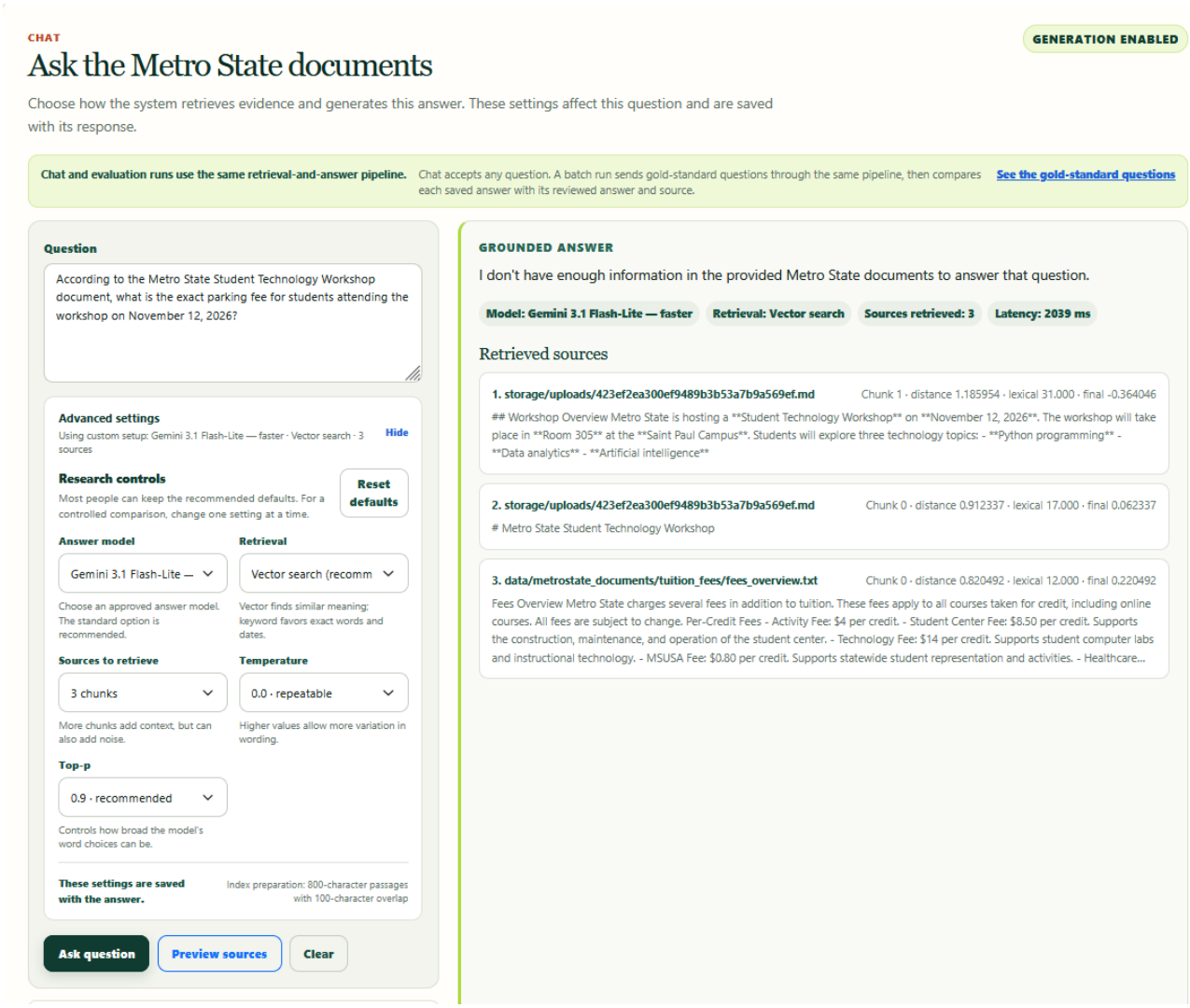


Figure 7.5 — Test Case 7 Question 3 Answer Screenshot

MySQL Metadata Check

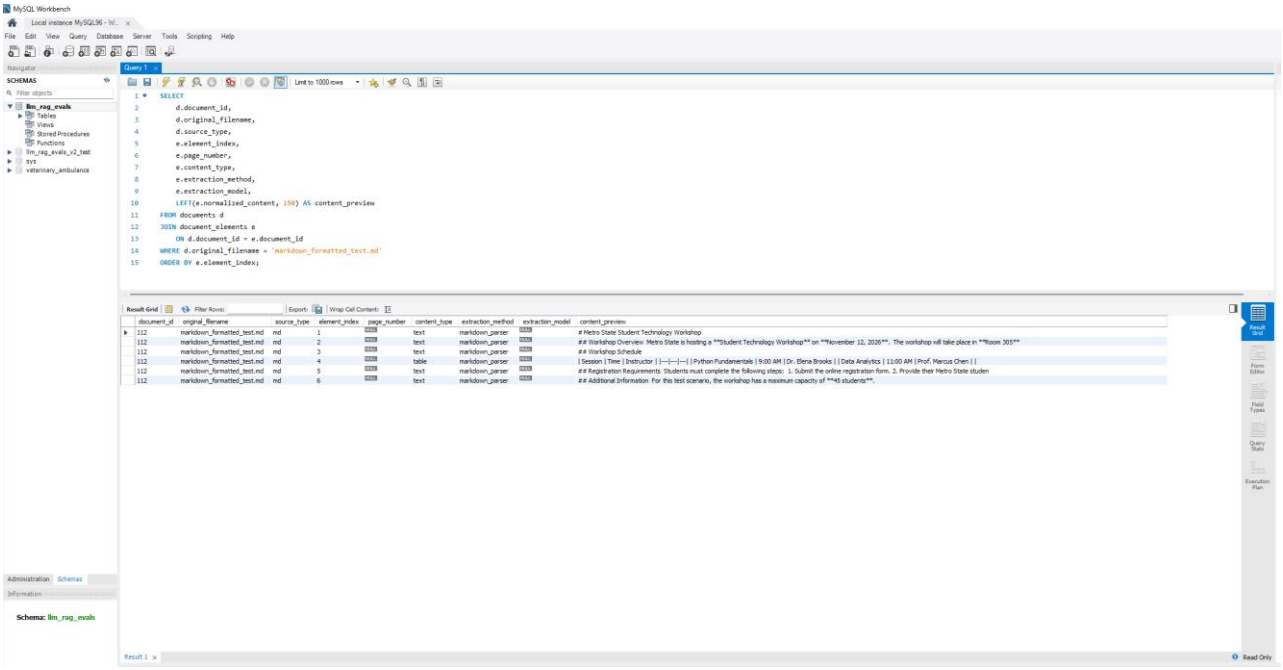


Figure 7.6 — Test Case 7 MySQL Metadata Check Screenshot

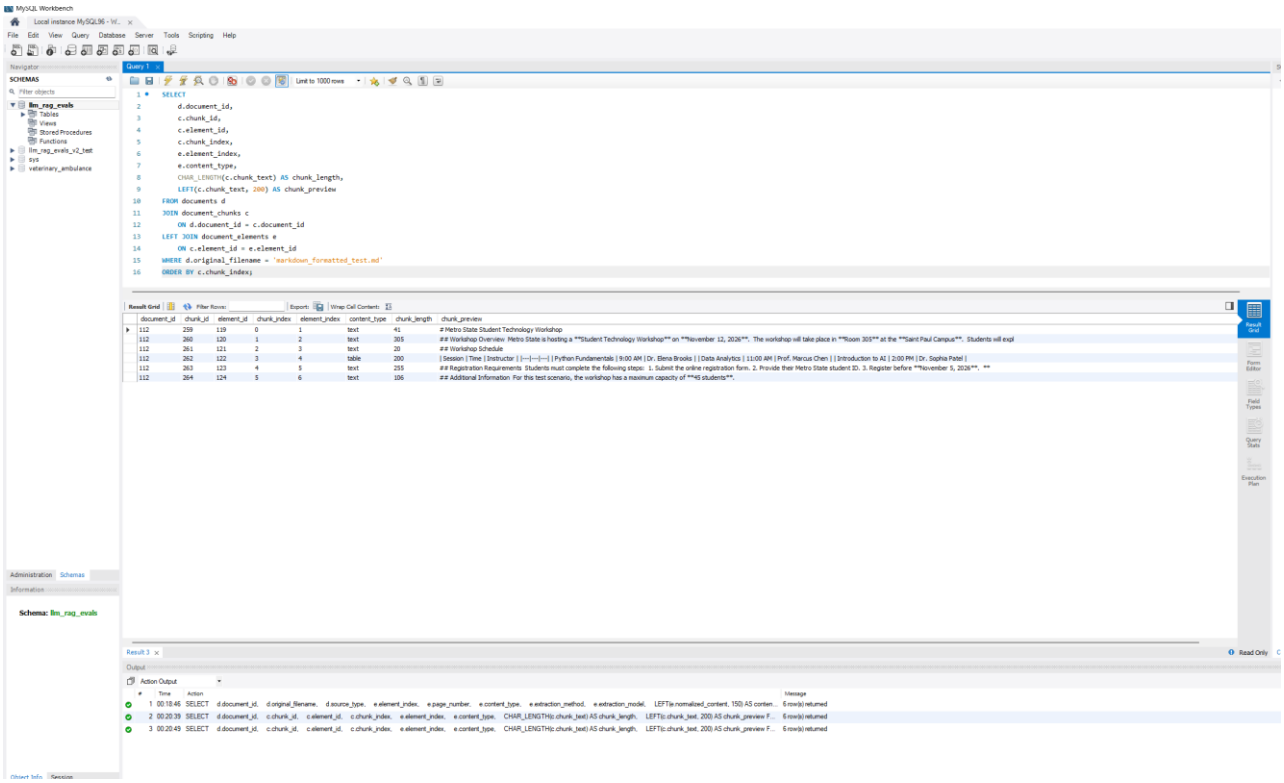


Figure 7.7 — Test Case 7 Markdown Chunk Content Screenshot

Assessment

PASS — supplied chat evidence includes correctly answered factual questions and an appropriately unanswerable parking-fee question.

Test Case 08 — Markdown with Mermaid diagram

Input fixture: TC08_markdown_mermaid_test.md | Open / download from GitHub

Mermaid diagram and hybrid retrieval improvement

Mermaid flowchart syntax is parsed into searchable descriptions of nodes, transitions, and decision paths while retaining the original diagram source. Hybrid RRF combines semantic vector results and keyword results to improve retrieval when questions reference exact diagram labels or relationships. The supplied notes record that a diagram previously ranked about 20th in vector retrieval and subsequently reached the top three using the improved hybrid pipeline.

Original test notes and visual evidence

Ingestion

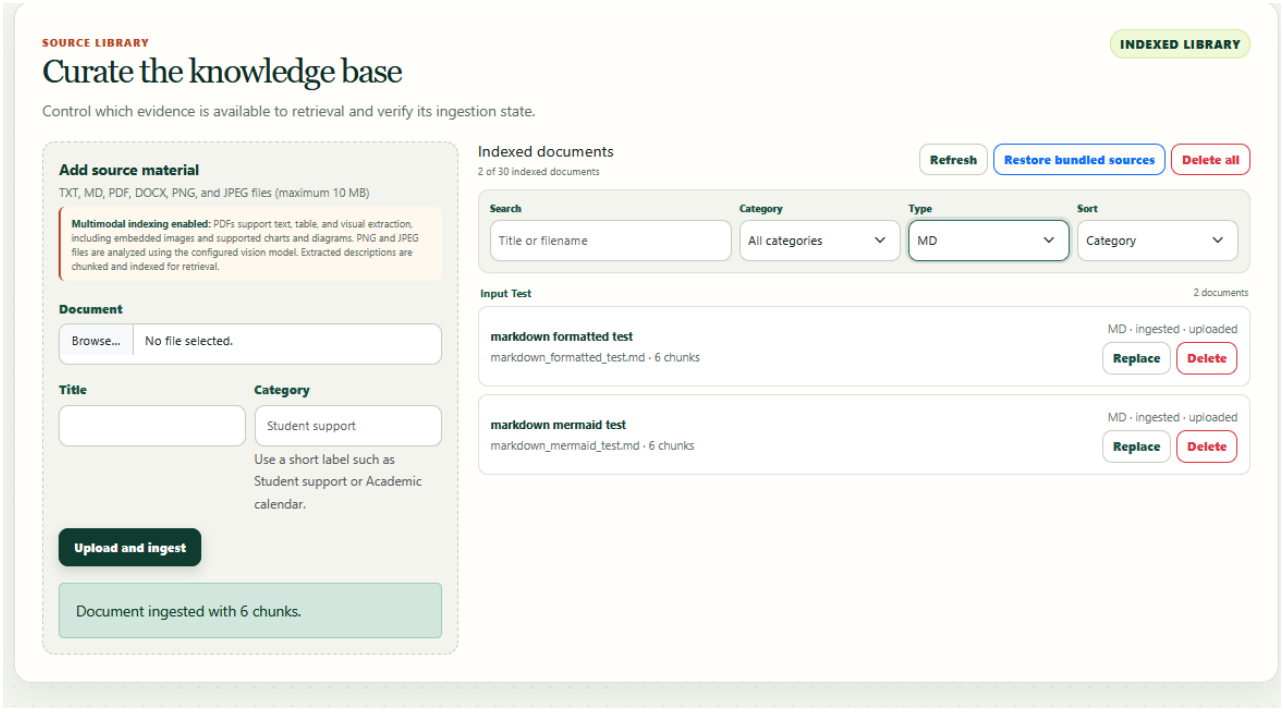


Figure 8.1 — Test Case 8 Document Ingestion Screenshot

RAG Retrieval and Chat Test

According to the Metro State Graduate Application Review Process document, what happens if an applicant has not submitted all required documents, and what happens after the student responds?

Expected: Missing documents are requested, the system waits for the student's response, and then checks document completeness again.

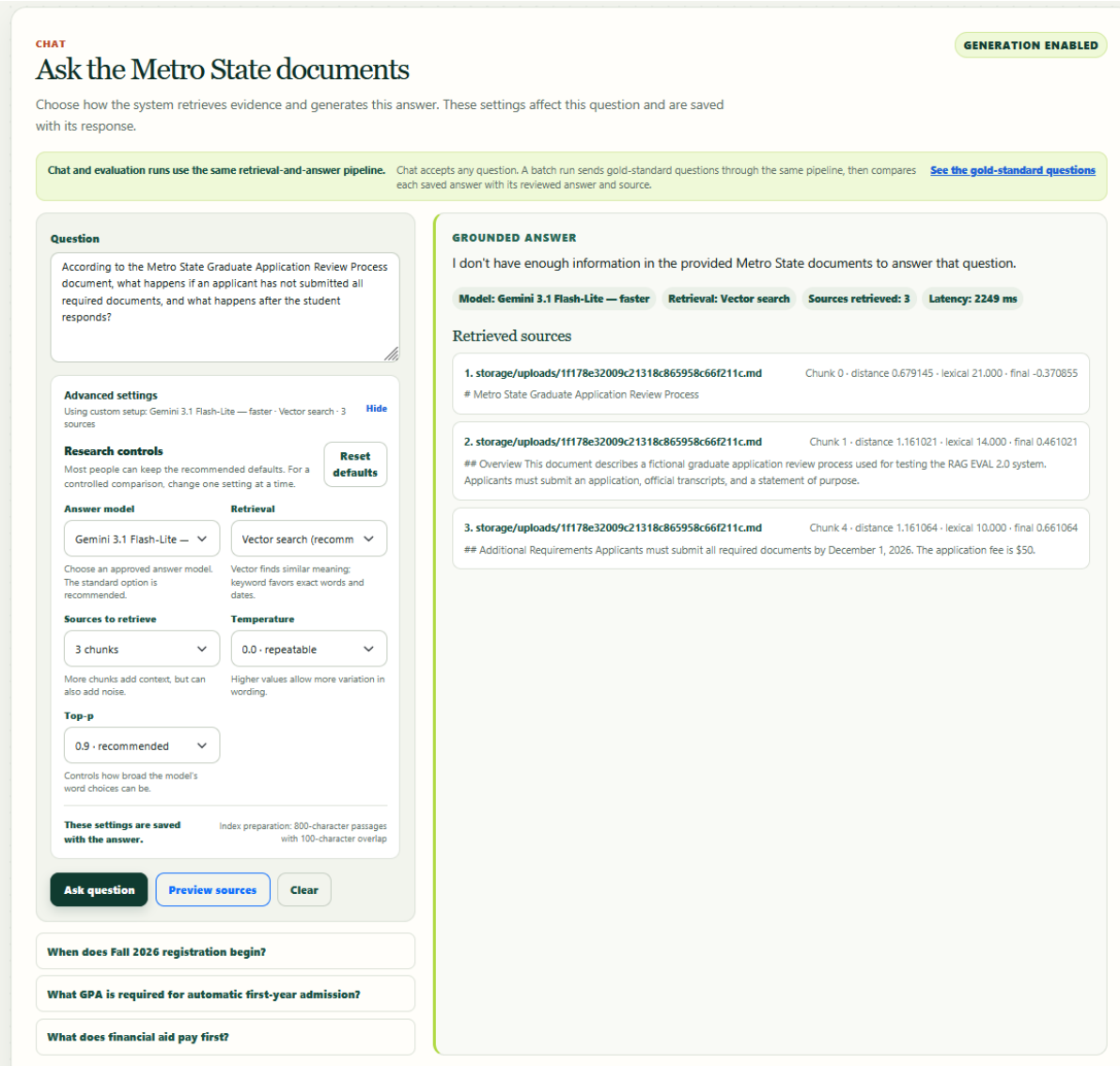


Figure 8.2 — Test Case 8 Question 1 Answer Screenshot

According to the Metro State Graduate Application Review Process diagram, what happens if an applicant has submitted all required documents but does not meet the minimum GPA of 3.0?

Copy

Expected: The application is rejected before reaching faculty committee review.

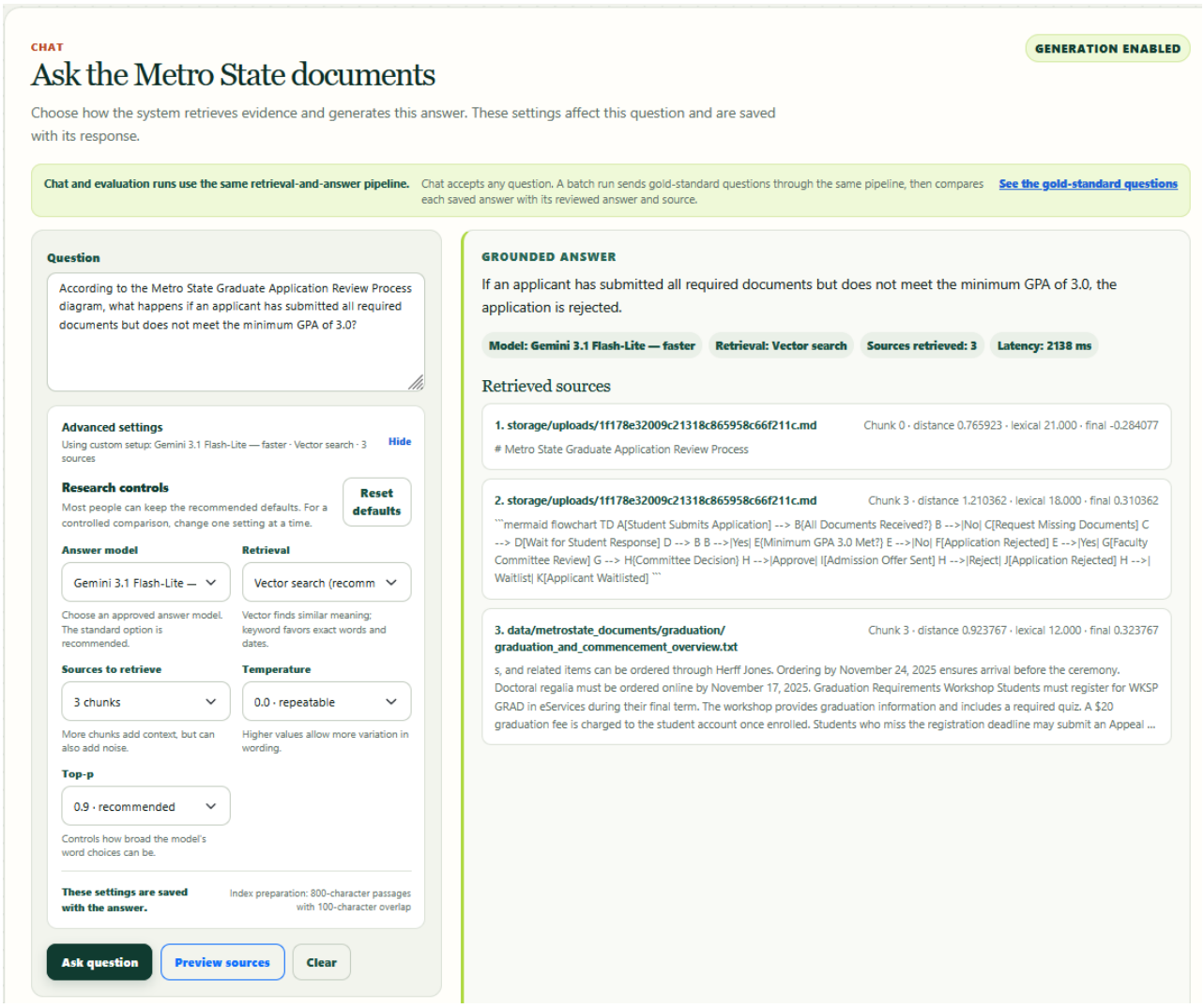


Figure 8.3 — Test Case 8 Question 2 Answer Screenshot

Using Hybrid search

Hybrid search is a retrieval method that combines two approaches to finding relevant information:

Vector search (ChromaDB): Finds information based on semantic meaning, even when the wording differs from the user's question.

Keyword search (MySQL): Finds information based on matching words and phrases from the question.

Our implementation uses Reciprocal Rank Fusion (RRF) to combine the rankings from both search methods and select the most relevant document chunks.

Why we implemented it

We implemented hybrid search because vector search alone was not consistently retrieving our Mermaid diagram when users asked questions about its workflow.

For our test question, the diagram originally ranked 20th in vector search, meaning it was excluded when retrieving only the top three chunks.

After improving the diagram's searchable descriptions and implementing hybrid search, the diagram ranked 3rd, allowing Gemini to use its relationships to generate the correct answer.

The **Reciprocal Rank Fusion (RRF)** formula is:

$$\boxed{\text{RRF}(d) = \sum_{r \in R} \frac{1}{k + \text{rank}_r(d)}}$$

Where:

- d = A document chunk being evaluated.
- R = The set of retrieval methods (ChromaDB vector search and MySQL keyword search).
- k = A smoothing constant, typically **60**.
- $\text{rank}_r(d)$ = The position of chunk d in retrieval method r's results (1st, 2nd, 3rd, etc.).

Example from RAG EVAL 2.0

Suppose our Mermaid diagram chunk ranks:

- **8th** in ChromaDB vector search.
- **1st** in MySQL keyword search.

With $k = 60$:

$$\text{RRF}(d) = \frac{1}{60 + 8} + \frac{1}{60 + 1}$$

$$= 0.014706 + 0.016393$$

$$\boxed{\text{RRF}(d) \approx 0.031099}$$

Figure 8.4 — Reciprocal Rank Fusion (RRF) Formula and Example

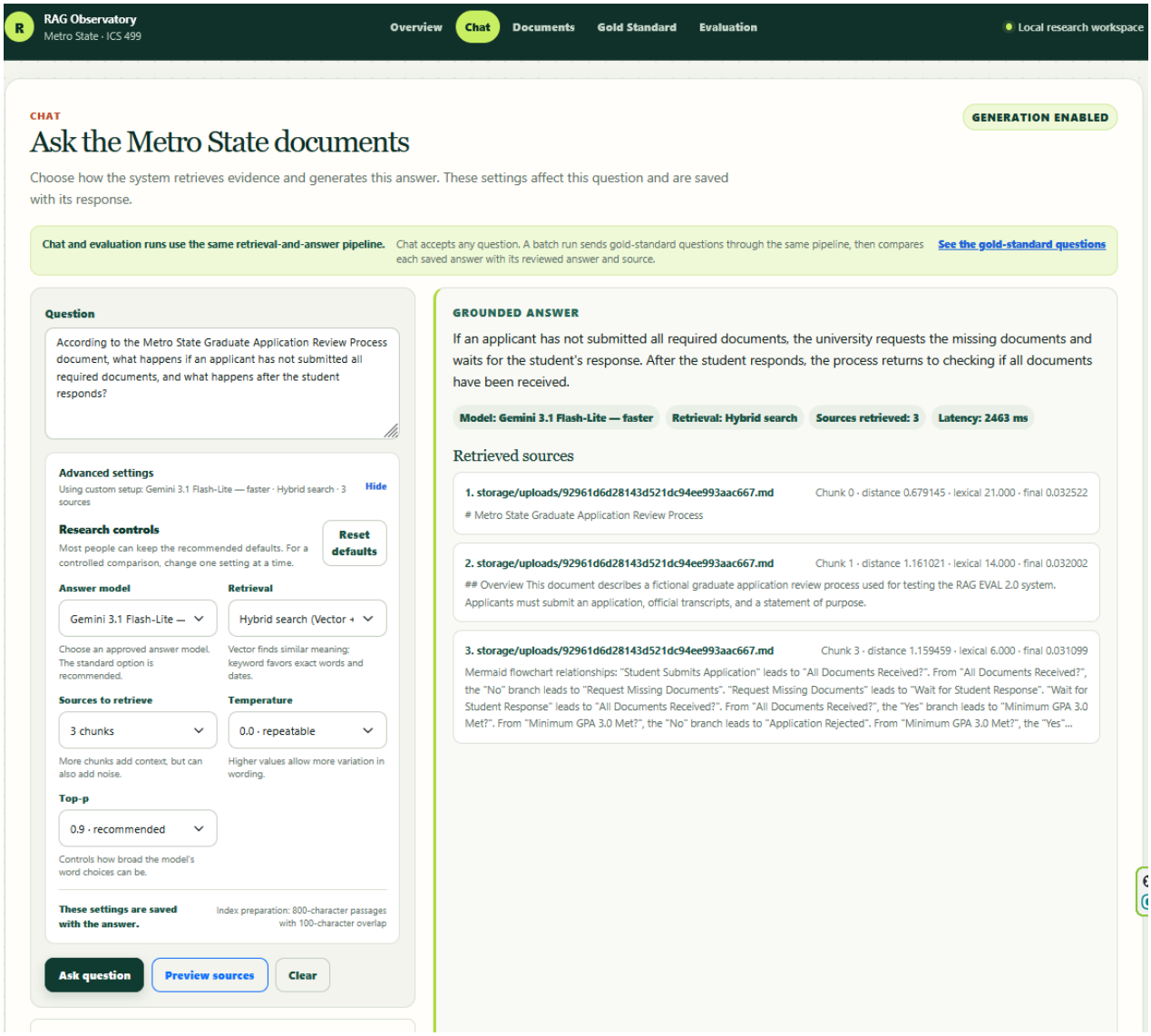


Figure 8.5 — Test Case 8 Hybrid Retrieval Evidence Screenshot

MySQL metadata check

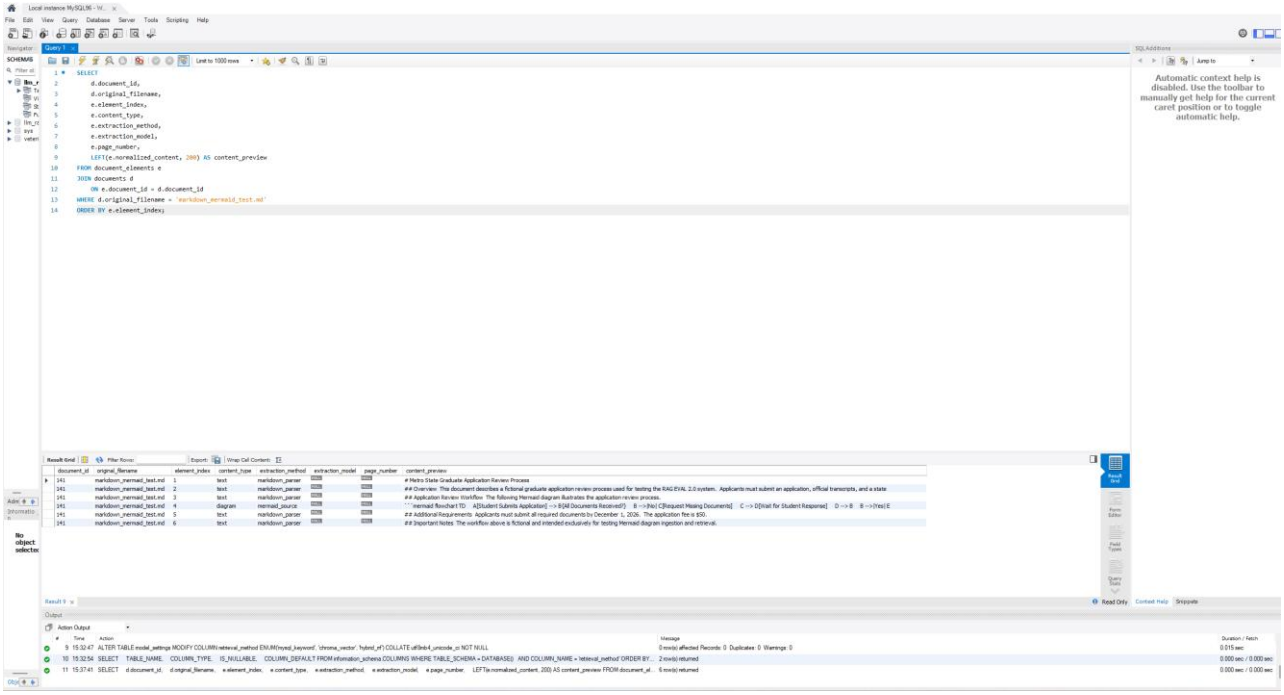


Figure 8.6 — Test Case 8 MySQL Metadata Check Screenshot

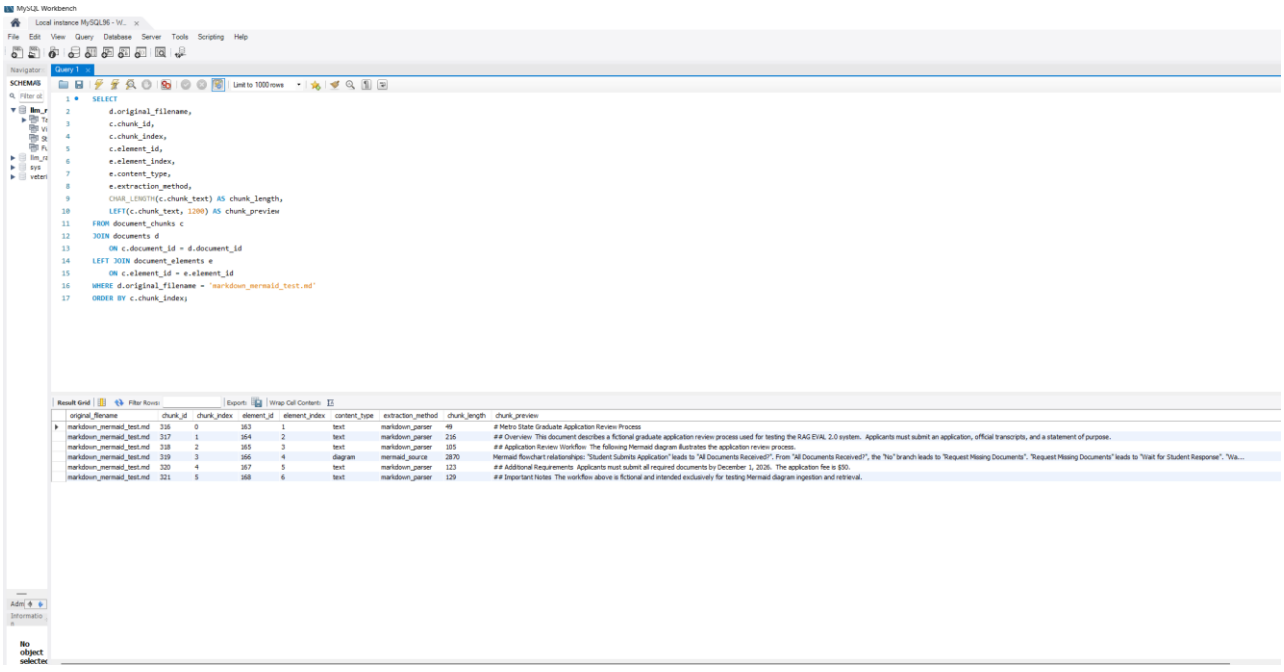


Figure 8.7 — Test Case 8 Mermaid Diagram Metadata Screenshot 1

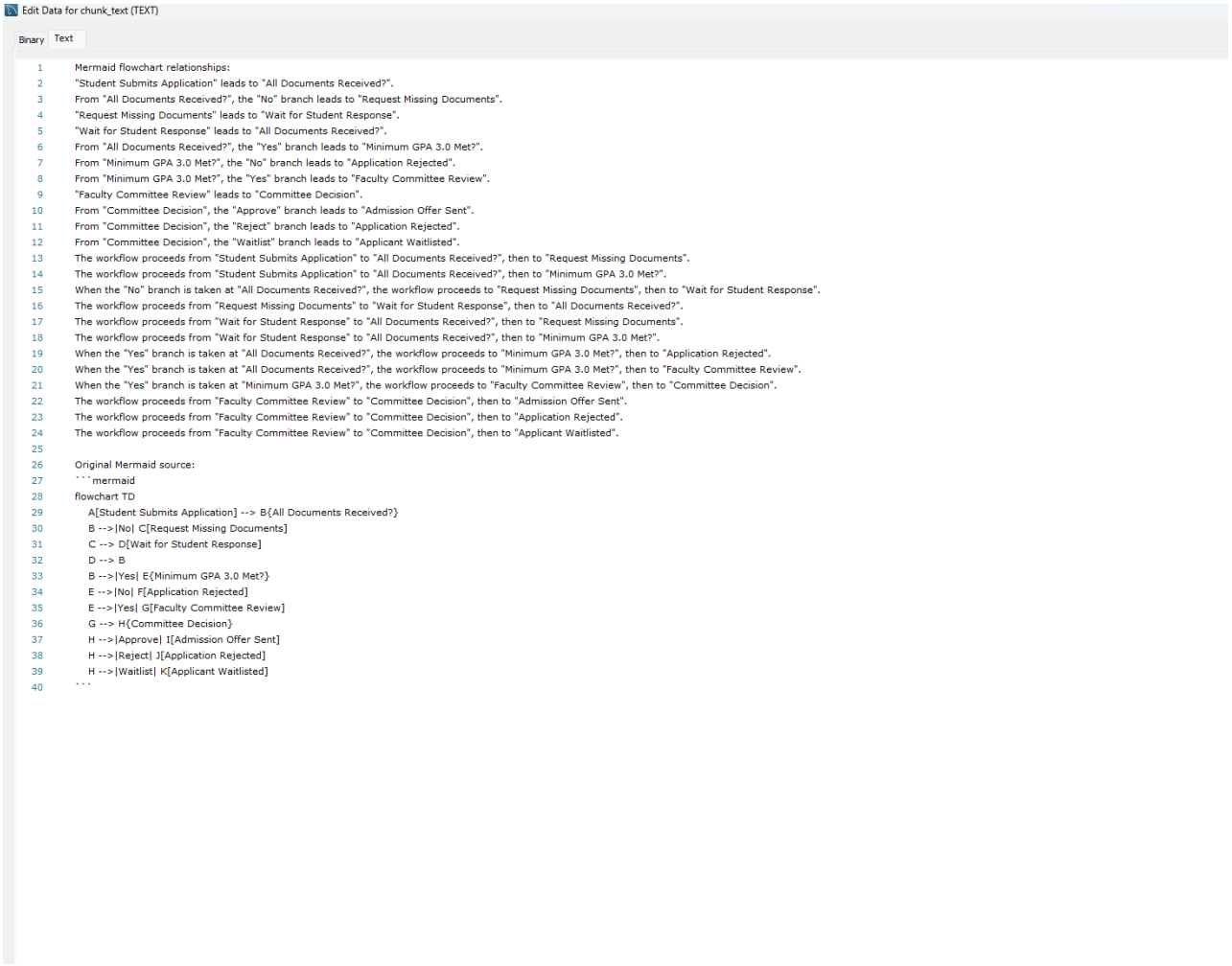


Figure 8.8 — Test Case 8 Mermaid Diagram Metadata Screenshot 2

Assessment

PASS — supplied evidence documents Mermaid relationship retrieval and correct grounded answers using the improved hybrid pipeline.

Test Case 09 — PowerPoint with text, tables, and images

Input fixture: TC09_multimodal.pptx | Open / download from GitHub

Original test notes and visual evidence

Ingestion

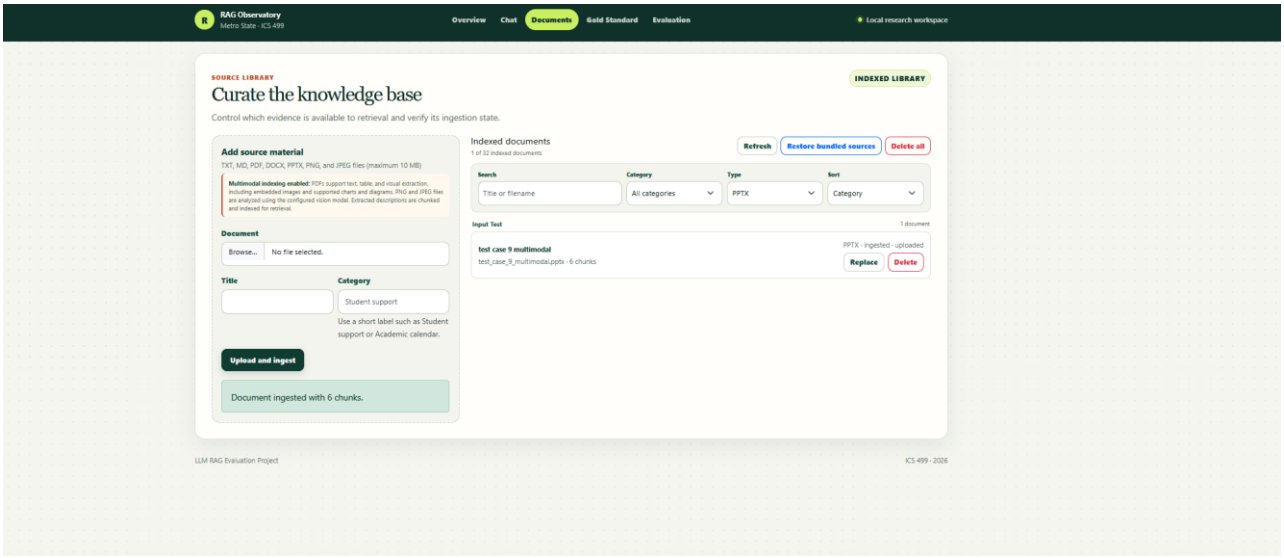


Figure 9.1 — Test Case 9 Document Ingestion Screenshot

MySQL Metadata Check

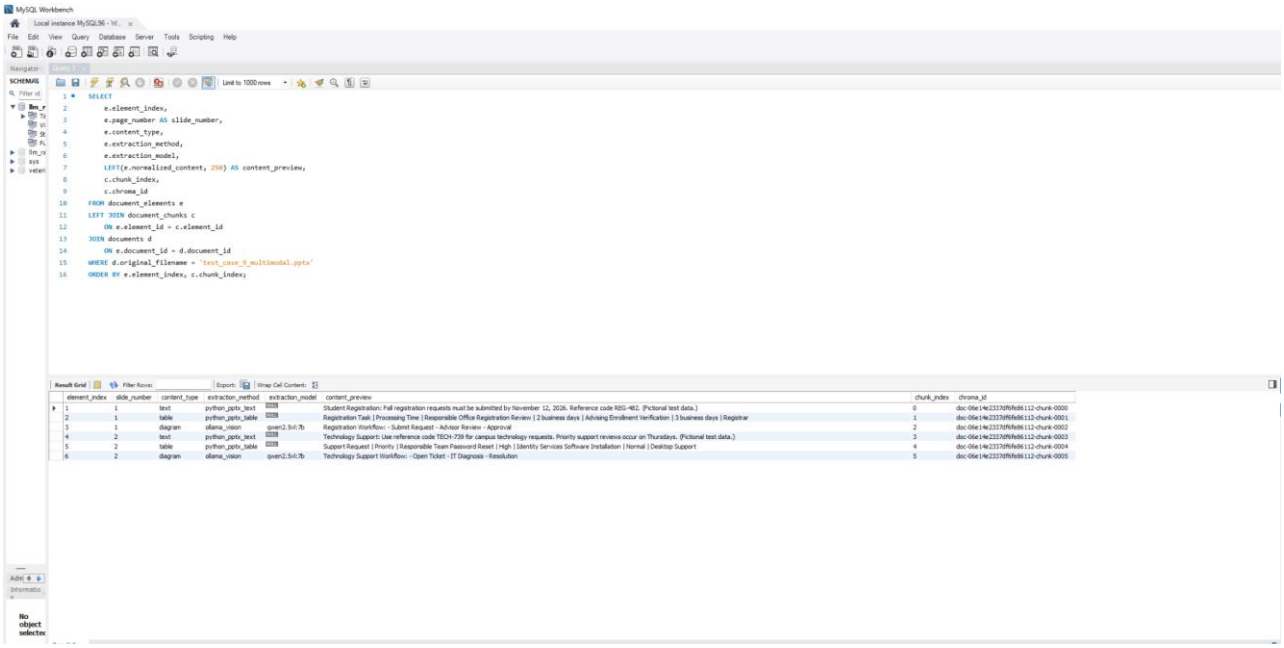


Figure 9.2 — Test Case 9 MySQL Metadata Check Screenshot

Chat

Slide text: What is the Fall 2026 student registration deadline and reference code according to the PowerPoint presentation?

Expected: November 12, 2026; REG-482.

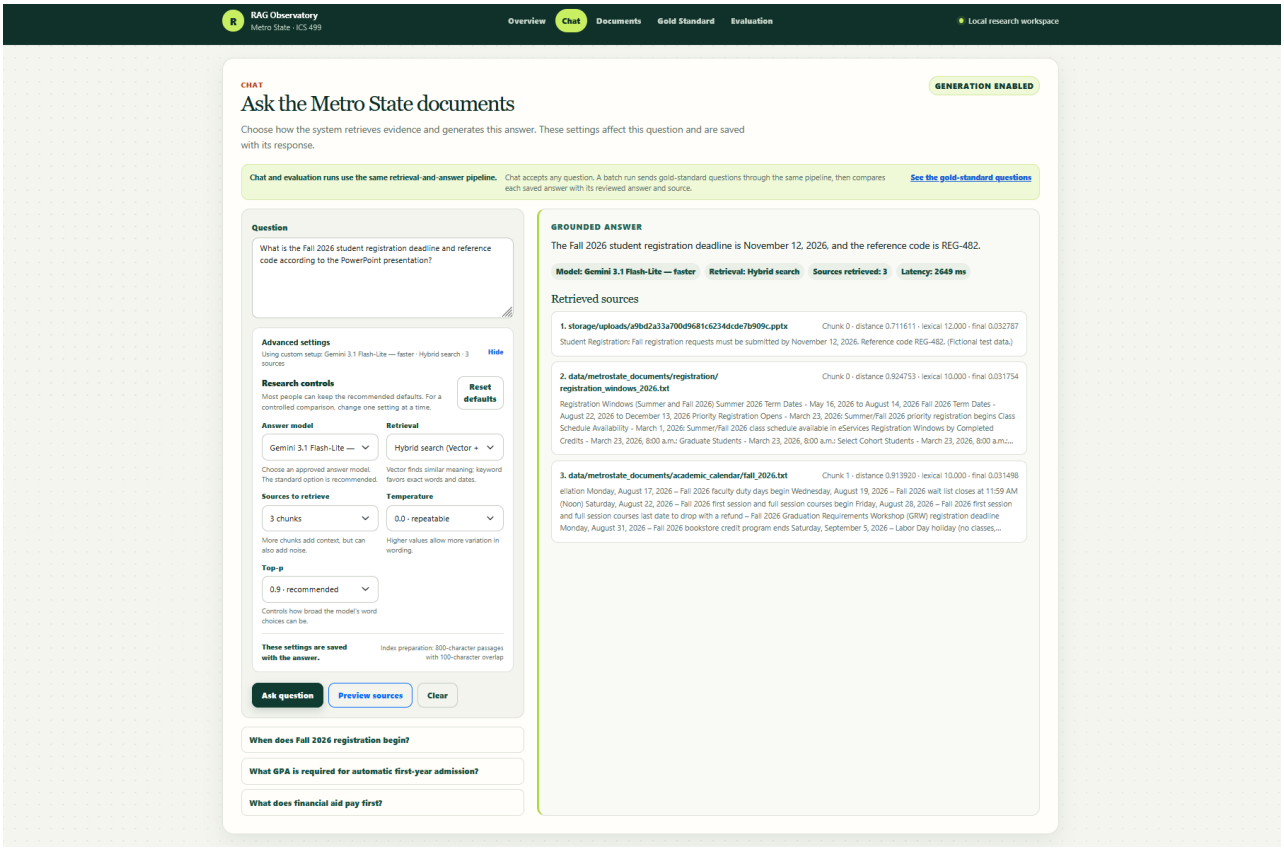


Figure 9.3 — Test Case 9 Question 1 Slide Text Answer Screenshot

PowerPoint table: According to the PowerPoint registration task table, how long does Enrollment Verification take and which office is responsible?

Expected: 3 business days; Registrar.

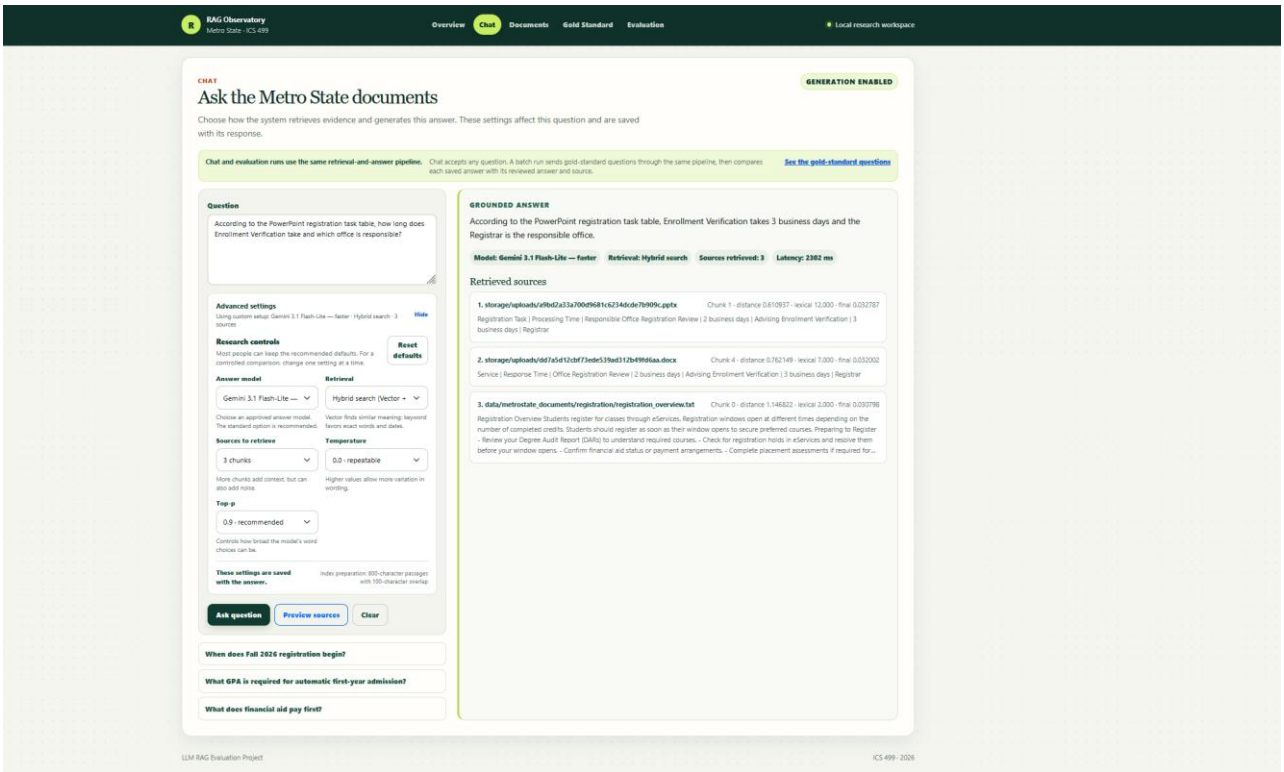


Figure 9.4 — Test Case 9 Question 2 Table Answer Screenshot

Slide Image/diagram: According to the PowerPoint Technology Support Workflow diagram, what are the three steps in order?

Expected: Open Ticket → IT Diagnosis → Resolution.

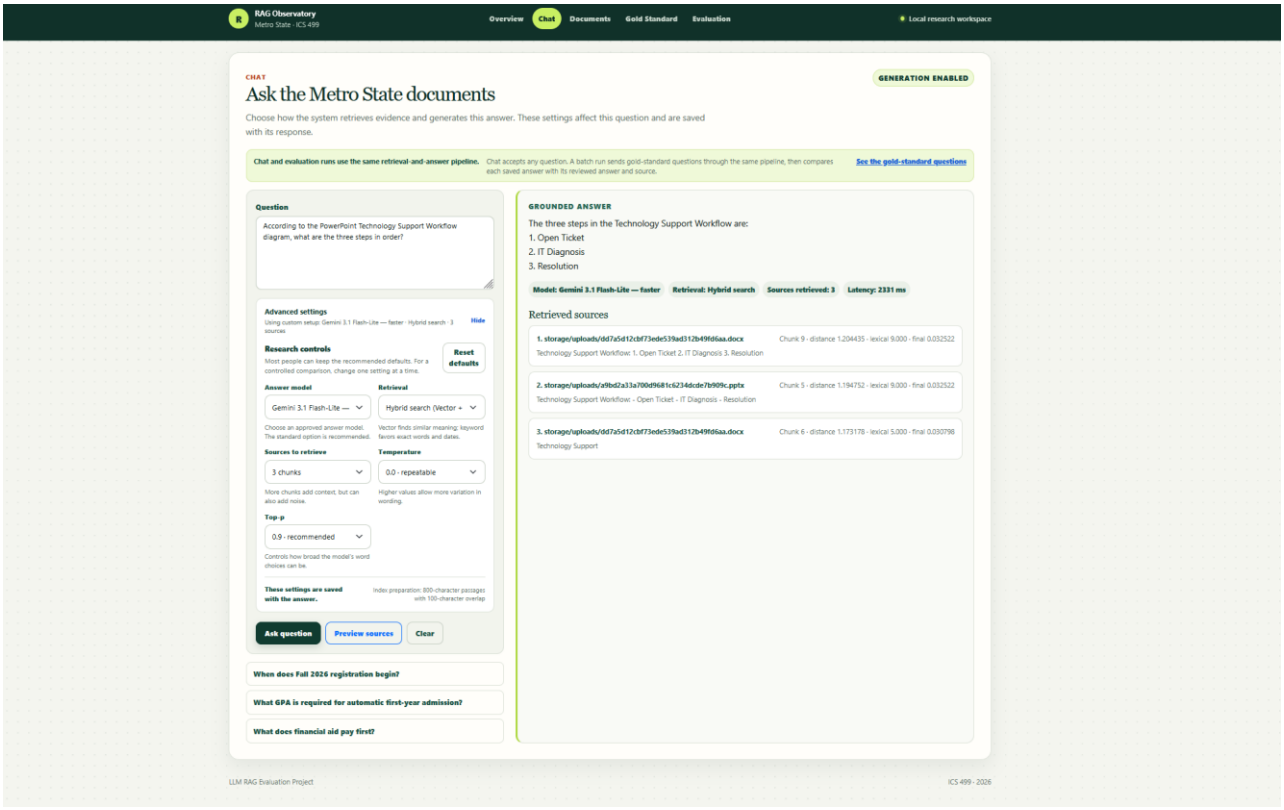


Figure 9.5 — Test Case 9 Question 3 Diagram Answer Screenshot

Assessment

PASS — supplied screenshots document ingestion and grounded answers matching the stated expected results. See individual figures and question/expected-answer notes above.

Test Case 10 — Word document with text, tables, and images

Input fixture: TC10_multimodal.docx | Open / download from GitHub

Original test notes and visual evidence

Ingestion

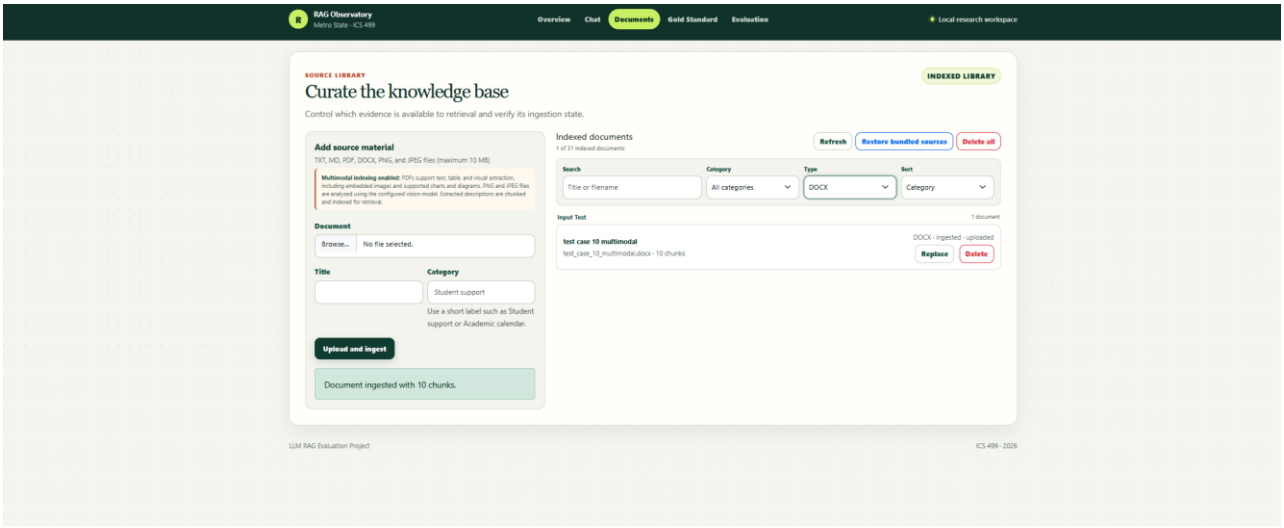


Figure 10.1 — Test Case 10 Document Ingestion Screenshot

MySQL Metadata check

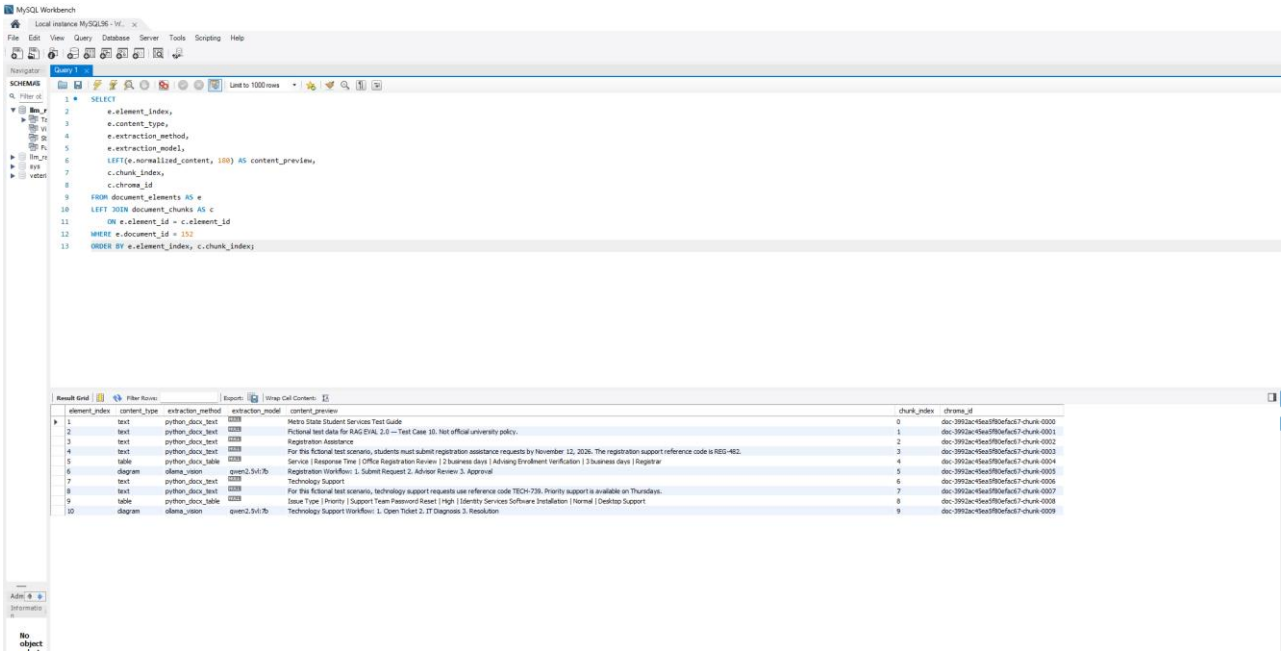


Figure 10.2 — Test Case 10 MySQL Metadata Check Screenshot

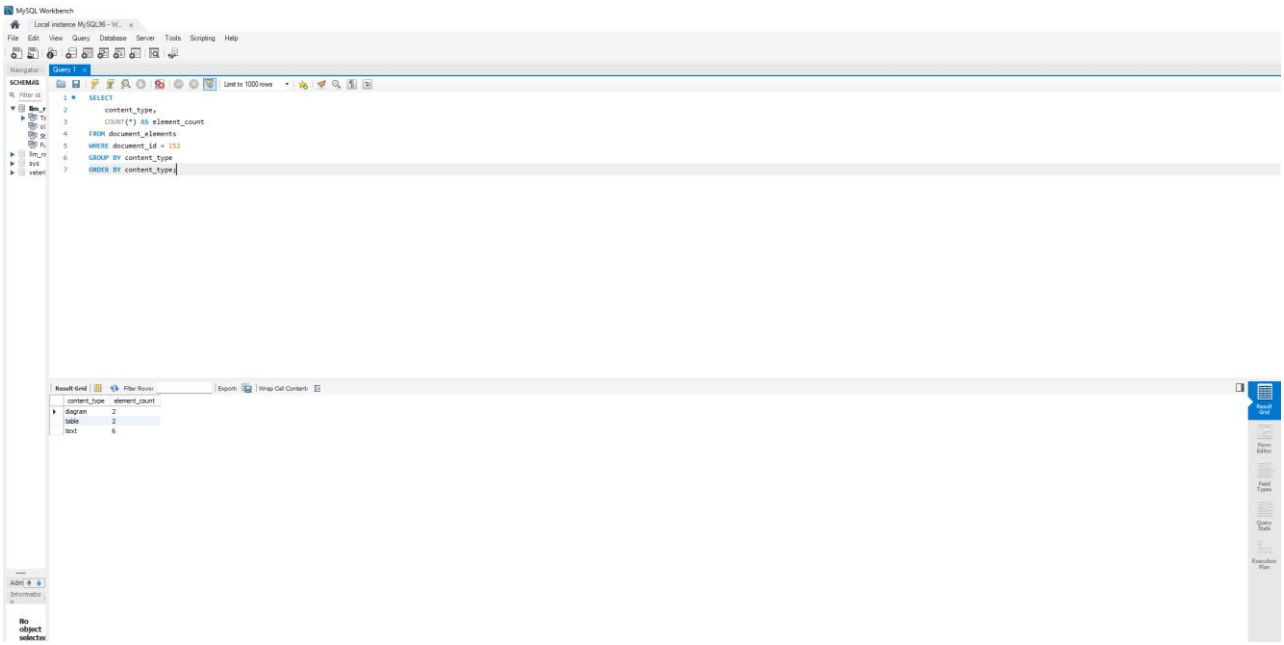


Figure 10.3 — Test Case 10 Document Chunk Content Screenshot

Chat

Text retrieval: According to the fictional Metro State Student Services Test Guide, what is the registration assistance deadline and reference code?

Expected answer: November 12, 2026; REG-482

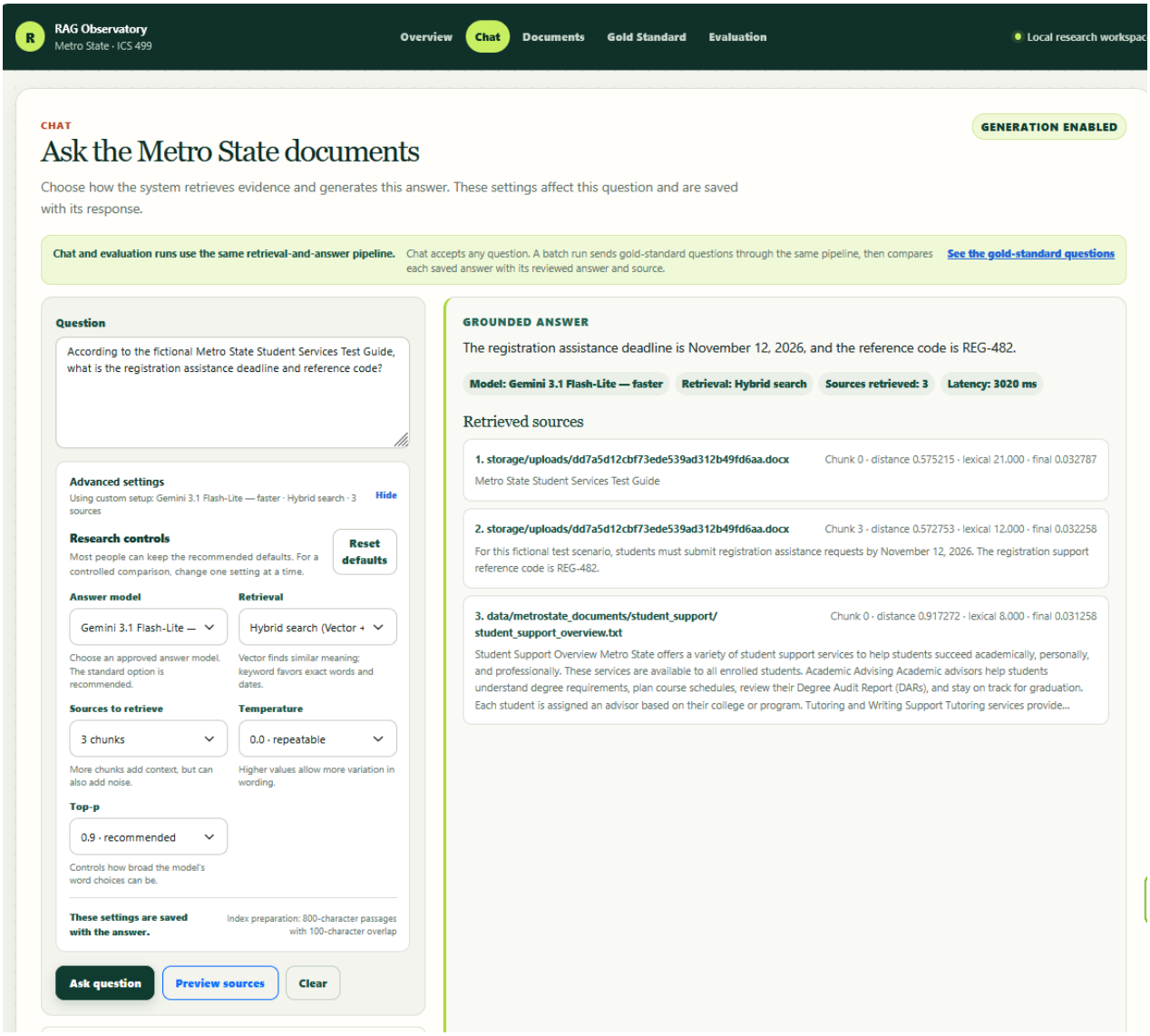


Figure 10.4 — Test Case 10 Question 1 Text Answer Screenshot

Table retrieval: According to the Student Services Test Guide, what is the response time and responsible office for Enrollment Verification?

Expected answer: 3 business days; Registrar

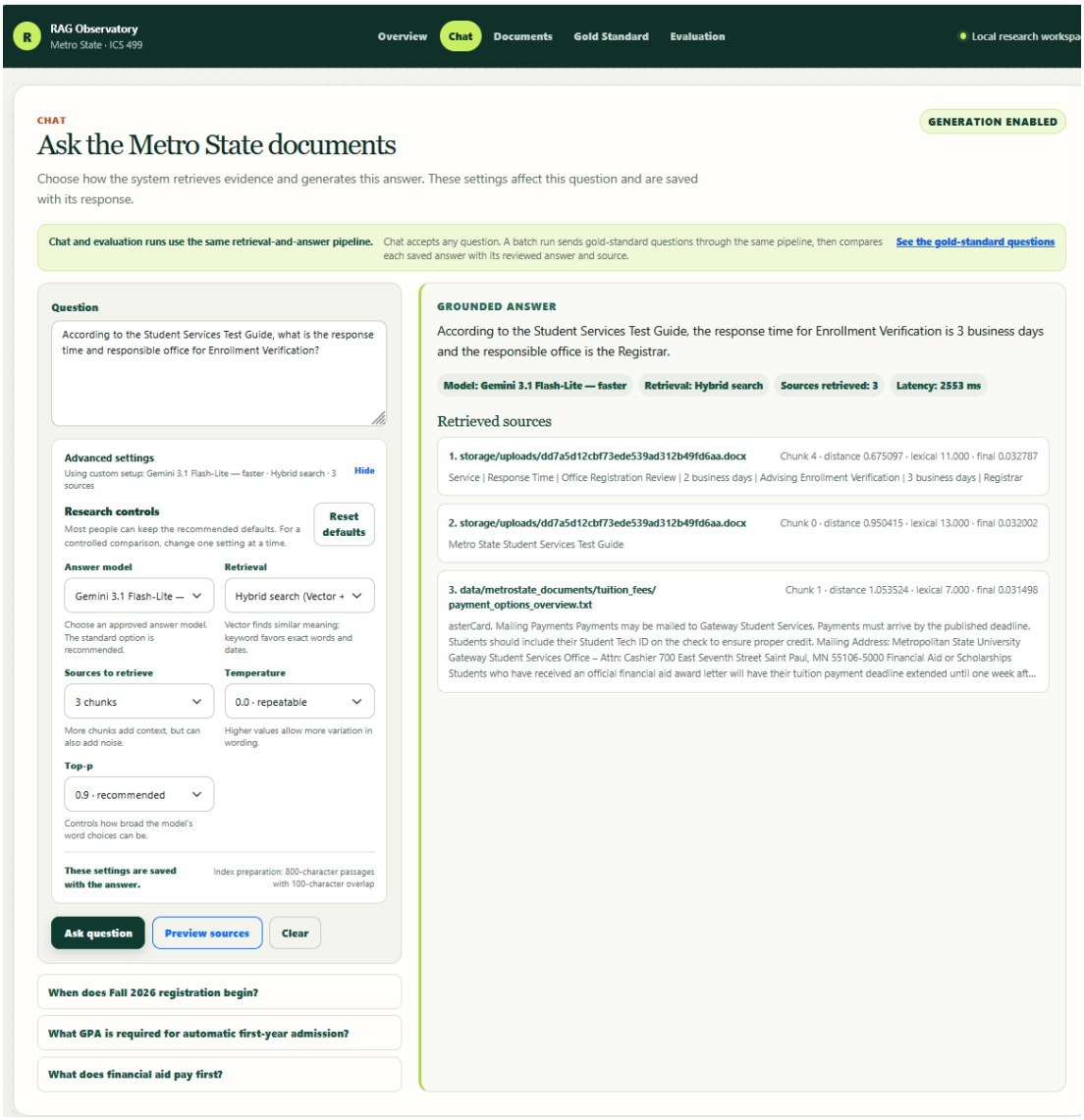


Figure 10.5 — Test Case 10 Question 2 Table Answer Screenshot

Image diagram retrieval: According to the Technology Support Workflow diagram in the Student Services Test Guide, what are the three steps in order?

Expected answer: Open Ticket → IT Diagnosis → Resolution

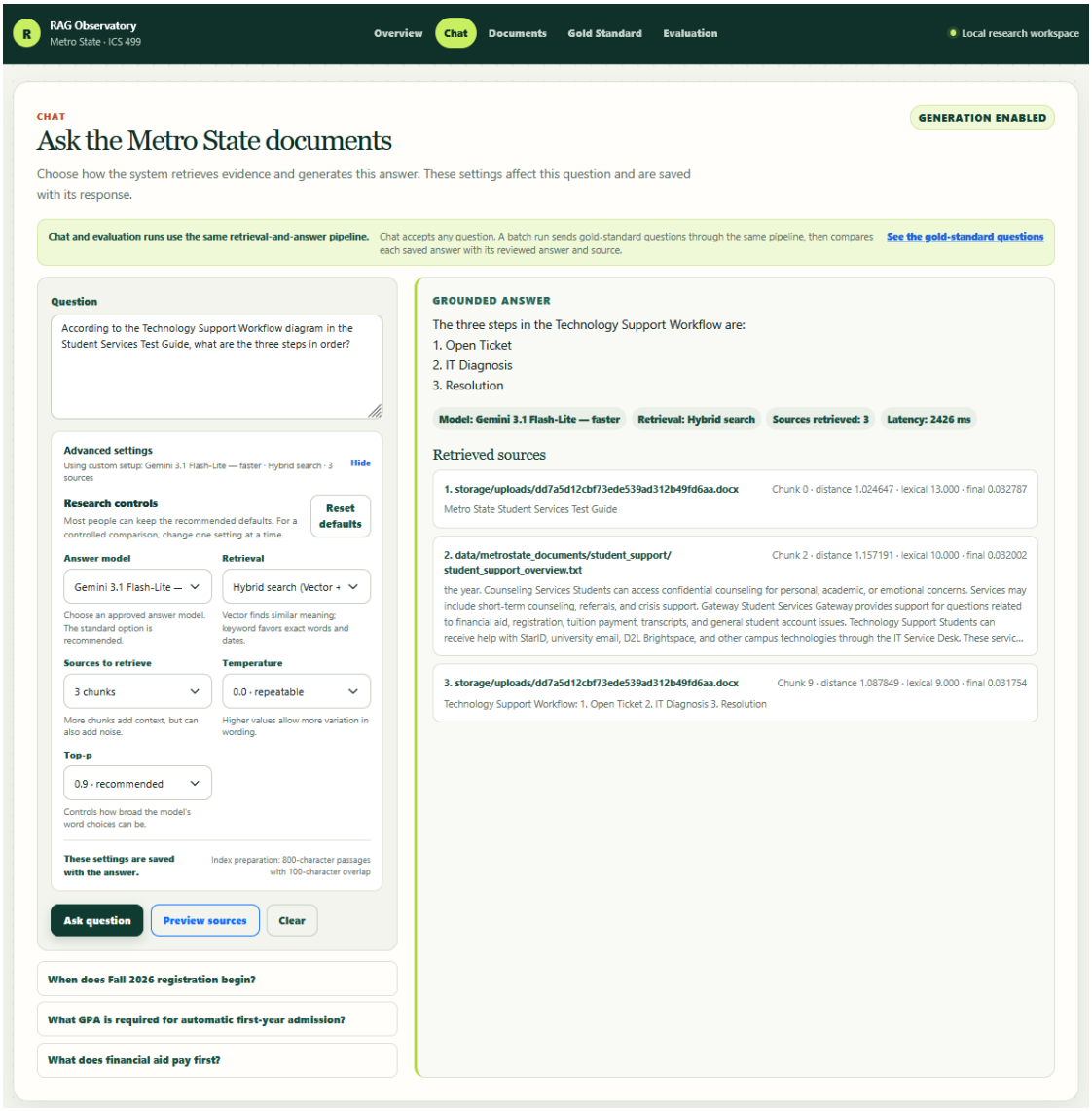


Figure 10.6 — Test Case 10 Question 3 Diagram Answer Screenshot

Assessment

PASS — supplied screenshots document ingestion and grounded answers matching the stated expected results. See individual figures and question/expected-answer notes above.

Conclusions

The ten test fixtures and accompanying evidence demonstrate multimodal ingestion and grounded question answering across the requested input formats. Test Case 6 highlights a practical retrieval limitation: relevant structured or visual evidence may exist in the index but fall outside a small top-k selection. Natural-language table enrichment and content-aware reranking improved this behavior without changing the stored original source content.

Future work includes reducing duplicate visual extraction, adding explicit reranking metadata to evaluation-run configurations, and extending controlled benchmarking across a larger question set. These items are not presented as completed enhancements.

Implementation and fixture commit eae5e9b